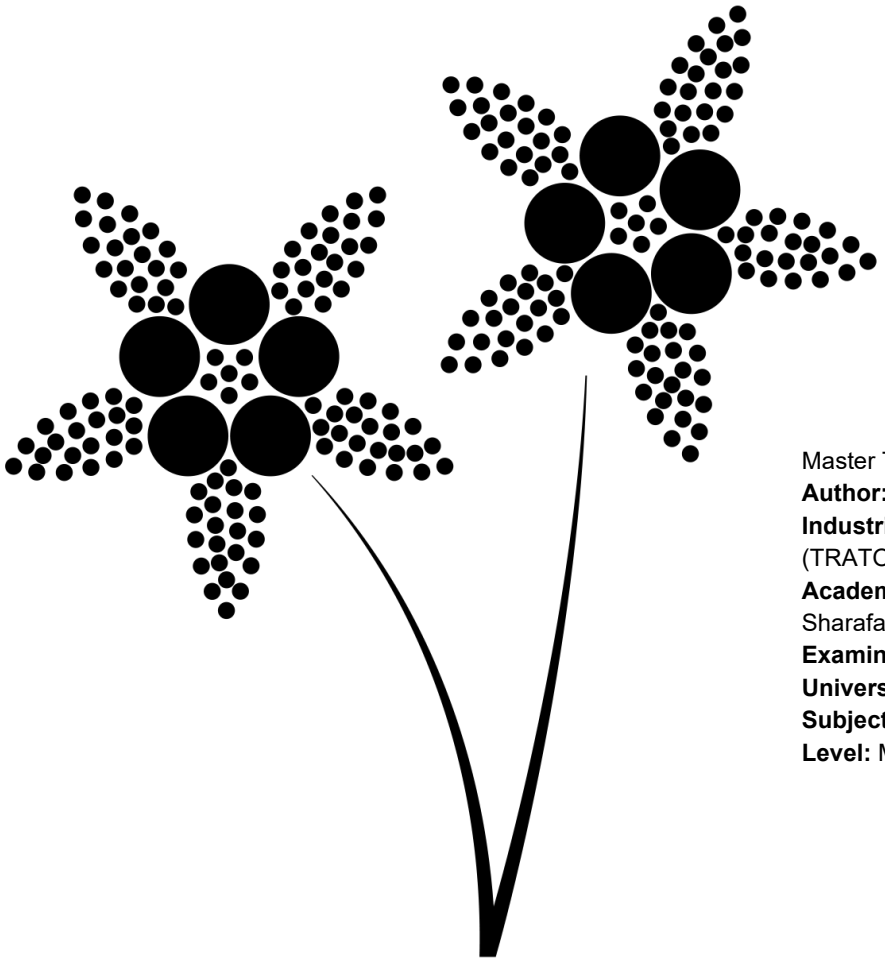
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Assessment of disassembly strategies and circularity potential of permanent magnets in electric vehicle applications



Master Thesis Report
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Level: Master's level

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Abstract

The rapid expansion of electric vehicles (EVs) is intensifying demand for high-performance Nd-Fe-B permanent magnets, which rely on critical rare-earth elements (REEs) such as neodymium and dysprosium. This dependence presents environmental, economic, and geopolitical challenges due to resource-intensive extraction processes and concentrated global supply chains. Improving circularity through efficient end-of-life (EoL) recovery of permanent magnets is therefore essential.

This thesis investigates the feasibility of hydrogen decrepitation (HD) as a short-loop recycling strategy for recovering Nd-Fe-B magnets embedded in EV rotor assemblies. A combined methodology is applied, integrating literature review, environmental impact assessment, and experimental evaluation. The environmental analysis highlights significant impacts associated with virgin magnet production, particularly driven by REE extraction and processing, reinforcing the importance of recycling strategies.

Experimental work focuses on hydrogen-assisted disassembly of magnets within rotor stacks under controlled temperature and pressure conditions. Results show that HD effectiveness is strongly dependent on surface accessibility. Magnets that presented intact metallic coatings exhibited no reaction, while coating removal enabled hydrogenation, volume expansion, and material fragmentation. However, complete pulverization was not achieved under the applied conditions.

The findings demonstrate that while hydrogen decrepitation has a strong potential as an energy-efficient and environmentally favourable recycling route, its direct application to embedded magnets remains constrained by material design and process conditions. The study identifies critical factors influencing feasibility, including coating type, surface exposure, and parameter optimization.

A decision-support framework is proposed to guide industrial selection of recycling pathways based on technical feasibility, circularity potential, and material characteristics aligned with business priorities. Overall, this work contributes to advancing sustainable strategies for permanent magnet recovery, supporting EU critical raw material goals and promoting circularity in EV supply chains.

Keywords

Permanent magnet, NdFeB, Disassembly, Recycling, Circularity, Hydrogen, Electric Vehicle, Decision-making framework.

Acknowledgments

I would like to thank TRATON Group and Linnaeus University for the opportunity of working on this project. To my supervisors Xiao Xiong (Research Engineer at TRATON Group R&D) and Sharafat Ali (Associate prof. at Linnaeus University) my most sincere gratitude for all the support and enlightenment you have provided through this thesis. To my family and friends that encouraged me to pursue my studies and that helped me to persevere through moments of hardship. To all of you, my deepest and sincerest thanks!

List of acronyms

EoL – End of life

EU – European Union

EV – Electric vehicle

HD – Hydrogen decrepitation

HDDR – Hydrogenation, Disproportionation, Desorption, and Recombination

HREE – Heavy rare-earth element

GB – Grain boundary

GBD – Grain boundary diffusion

IAC – Ionic adsorption clays

LREE – Light rare-earth element

LREO – Light rare-earth oxide

PM – Permanent magnet

RE – Rare-earth

REE – Rare-earth element

REM – Rare-earth mineral

REO – Rare-earth oxide

List of specific materials

Al – Aluminum

B – Boron

Co – Cobalt

Cu – Copper

Dy – Dysprosium

Dy_2O_3 – Dysprosium oxide

Fe – Iron

Ga – Gallium

Nd – Neodymium

Nd_2O_3 – Neodymium oxide

Nd-Fe-B PM – Neodymium-iron-boron permanent magnet

Nd(Dy)FeB PM – Dysprosium-doped neodymium-iron-boron permanent magnet

Ni – Nickel

Pr - Praseodymium

Tb – Terbium

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1 Introduction

1.1 Background & Motivation

Climate change is currently one of the world's most pressing challenges, with impacts spanning multiple sectors. It affects economic activities such as agriculture and manufacturing, generates wide-ranging social consequences, and drives environmental degradation, including rising temperatures, land and water degradation, biodiversity loss, and an increasing frequency of extreme events [1]. A sustainable energy transition is therefore essential to reduce anthropogenic impacts associated with fossil fuel use, which accounts for approximately 90% of global CO_2 emissions. Of this total, the transport sector alone contributes more than one third [2], [3], [4]. Electrification in the automotive industry is a key step in the energetic transition towards decarbonization and long-term sustainability [5]. The expansion of electrification reduces oil demand and drives its displacement, which exceeded 1.3 million barrels per day in 2024 [6].

Modern EV powertrains rely on battery systems and high-efficiency traction machines, where permanent magnets (PMs) are often used as critical components in permanent-magnet synchronous motors (PMSMs). These motors are widely used because they provide high efficiency, high torque density, compact design, and good controllability, which are essential for high-performance EV applications. The PMs enable efficient electromechanical energy conversion, enhance motor performance, and allow for compact powertrain designs. High-performance NdFeB magnets provide high remanence and maximum energy product, enabling stronger magnetic flux density and improved torque generation in compact motor geometries. These permanent magnets for EVs are typically made from a combination of transition metals, such as iron (Fe), and rare-earth elements, such as neodymium (Nd), dysprosium (Dy), and terbium (Tb). However, this reliance poses significant challenges related to resource availability and the environmental impacts of rare-earth mining [7].

Rare-earth elements (REEs) is a group comprised of 17 chemical elements, including yttrium, scandium, and the 15 lanthanides on the periodic table. These elements can be characterized for having an inert gas core, any quantity of "f-orbital" electrons, and three main valence electrons. They are metals that often present similar properties and can frequently be found together in geologic deposits in the form of oxides, with rare-earth mineral deposits being mostly found in China, USA, India, Australia, Brazil, Russia, Vietnam, and Myanmar. Although REEs are not necessarily scarce in absolute geological abundance, the term "rare earth" reflects the difficulty of finding, extracting, separating, and refining them economically, since they are often dispersed in minerals at relatively low concentrations. Their production requires multiple processing steps, including mining, beneficiation, separation, chemical treatment, and solvent extraction, before individual rare-earth compounds or metals can be obtained. Because of their exceptional magnetic, catalytic, and optical properties, REEs are indispensable for several clean-energy, digital,

and strategic technologies, including EVs, wind turbines, electronics, medical devices, and defence-related applications [8], [9]. The Raw Material Information System (RMIS) therefore defines REEs as essential for the maintenance and advancement of green and digital transitions, besides defence and aerospace industries, and classifies them as strategic and critical raw materials [10].

The European Union (EU) is heavily dependent on imports of critical raw materials. This dependency is particularly important for PMs, where the supply chain is highly concentrated outside Europe. China plays a dominant role in rare-earth mining, refining, and permanent magnet manufacturing, which creates strategic vulnerability for European industries [11], [12]. This supply concentration exposes European value chains to geopolitical, economic, and trade-related disruptions. In response, the EU has increased its focus on securing a sustainable and resilient supply of critical raw materials by strengthening domestic capacity, diversifying supply, and promoting recycling and circularity [13].

In 2024, the EU adopted the Critical Raw Materials (CRMs) Act, which establishes strategic raw materials and sets benchmarks to be achieved by 2030 [14]. For strategic raw materials, the CRMs Act aims to ensure that [14]:

- the EU can extract at least 10% of its annual consumption from within the Union;
- the EU can process at least 40% of its annual consumption, including intermediate processing steps;
- the EU can produce at least 25% of its annual consumption through recycling, including intermediate recycling steps.

In addition to economic and supply-chain concerns, REE extraction and processing are associated with significant environmental and social impacts. These include land disturbance, water and soil pollution, high energy and chemical consumption, hazardous waste generation, and potential exposure to radioactive elements such as thorium and uranium in some rare-earth ores [15]. REE mining may also affect local communities through displacement, health risks, disruption of traditional livelihoods, and environmental justice concerns. In some regions, opposition from local communities has increased the complexity of developing new rare-earth mining projects [16], [17]. Therefore, improving the circularity of REE-containing products is not only a matter of resource efficiency, but also an important strategy for reducing environmental, social, and geopolitical pressures associated with primary REE production.

High performing permanent magnets, which have a lifecycle expectancy of 10 years, are a key component for continued advancement of the electrification in the transport sector and need special end-of-life (EoL) attention due to their strong REEs dependence [18]. Therefore, managing the EoL of PMs from EV rotors to achieve circularity is a critical step.

Current research focuses on recycling techniques for recovering REEs from EoL magnets, which is aligned to the necessity of addressing 300,000 tons of reported discarded REEs [19]. Projections from the International Energy Agency (IEA) indicate that by 2035, the stock of EVs will amount to approximately 505 million and electric trucks to 10 million [20]. A passenger EV rotor may contain 1- 2 kg up to 5 kg of PMs [21]. By correlating the data on rotor power output and rotor PM mass [21], and the power outputs from commercial heavy duty vehicles from TRATON Group [22], it is estimated that a heavy duty vehicle might require from 2.8 kg to over 10 kg of PMs depending on the required power output and the performance level of the magnets used .

According to IEA databases, global stock for EVs is projected to reach 250 million units by 2030, of which 3.2 million are expected to be trucks. At the vehicles' end of life (EoL), the quantity of permanent magnets would represent both a major recycling challenge and a significant opportunity for resource circularity [19], [20], [23]. Consequently, the development of efficient disassembly and PM recovery processes is essential to support sustainable transport electrification, reduce resource depletion, and mitigate supply-chain dependency. However, in contrast to the disassembly of magnets from other EoL applications such as wind turbine motors or hard drive disks, the magnets located in EV's rotor present obstacles that require further investigation.

In EV's, the magnets are subjected to high operating temperatures, vehicle's movement, and the high centrifugal force due to the rotors' rotation. Therefore, the magnets must be securely fixed within the rotor structure to prevent undesired movement or release during operation [24]. Considering manufacturers' increased reliance on polymeric adhesive bonding techniques to guarantee secure fixation of the PMs, understanding its impacts on the recovery methods chosen is also critical to magnet recovery [25]. In addition to the adhesive epoxy layer, magnets have a protective epoxy coating to prevent oxidation when exposed to atmospheric condition, oftentimes combined with a metallic alloy underlayer, increasing magnet protection but also recycling complexity [26].

Robotic disassembly of rotors follows a series of steps, such as optional demagnetization, disassembly of the bearings, fixations, and shaft, segmentation of the laminated stacks, and finally disassembly of PMs [24]. The mechanical disassembly of PMs requires considerations in relation to linear elastic deformation, adhesive damage and separation, and frictional behaviours. This kind of approach presents inherent challenges, such as dependence on the state of the EoL rotor – as damage and deformations hinder the robots' actions, influence of the type of adhesives used and friction in the required force, and potential damage to the magnet [24]. Robotic disassembly can then be said to face bottlenecks regarding material uncertainty. Alternative automated disassembly routes of permanent-magnet synchronous rotors have been studied as a possible pathway to address this bottleneck through adaptive automation and modelling [27]. However, there is much room for improvement in disassembly methods in industrial-scale, as a disassembly process is hard to automate [28].

1.2 Research gaps & Study relevance

Guaranteeing adequate disassembly and recovery of magnets in industrial scale enables circularity and reduces the strains caused by resource depletion and the impacts associated with REEs mining. To that end, critical evaluation must be made regarding disassembly and recovery processes associated with EV magnets. Disassembly and recovery are currently being investigated using various approaches, including physical, robotic, metallurgical separation methods, as well as hydrogen assisted processes [18], [29]. In view of these mechanical disassembly challenges, it is necessary to develop more energy-efficient, cost-effective, and low-damage disassembly routes for EV permanent magnets.

Besides robotic methods, it is important to pursue alternative permanent magnet disassembly routes. A central challenge addressed in this work is the possibility of recovering magnets without complete disassembly of the rotor structure. Several routes exist for recycling permanent magnets, including hydrogen processing of magnetic scrap (HPMS), hydrometallurgical treatment, pyrometallurgical treatment, ionic-liquid extraction, and molten salt electrolysis [30], [31], [32]. Among these, hydrogen-assisted methods are particularly relevant because they may combine magnet separation, demagnetization, and pulverization in a shorter recycling route.

At this moment, to the best of the author's knowledge, no studies have been performed on the use of hydrogen decrepitation as a integrated method for disassembly and recycling of embedded magnets in EV rotor stacks.

This thesis will provide relevant insights into hydrogen-assisted methods for disassembly and recycling of magnets embedded in EV rotor structures, based on clear environmental drivers. In addition, it will provide automotive business with structured frameworks to support the pursuit of material circularity and sustainable practices.

1.3 Aims & Objectives

This thesis therefore has the aim of mapping of process flow for the manufacturing of virgin Nd-Fe-B magnets and evaluating the associated the eco-costs. Additionally, it investigates hydrogen-assisted treatment as a potential route for disassembling and recovering PMs embedded in EV rotor structures. Finally, this study proposes decision frameworks to support the business in choosing the best pathways towards Nd-Fe-B magnet recovery and recycling.

1.4 Research questions

The following guiding questions are defined to drive objective achievement in this study:

- What are the key eco-costs and impacts associated with the manufacturing of new Nd-Fe-B magnets, and what are the main drivers of such impacts?

- Is hydrogen decrepitation a viable disassembly and circularity enabler for permanent magnets embedded in EV rotors?

- Which recycling pathway should industry prioritize under specific conditions of EoL EV NdFeB magnets?

1.5 Thesis structure

The thesis begins by defining the research problem and reviewing the current state of knowledge regarding Nd-Fe-B magnet production impacts, recycling technologies, and hydrogen decrepitation. Based on the identified knowledge gaps, a methodology was developed comprising three complementary research activities: environmental eco-cost assessment, experimental evaluation of hydrogen decrepitation for embedded magnets, and development of a recycling decision-support framework. The outputs of these activities form the basis of the results chapter, which are subsequently interpreted in the discussion and used to synthesize conclusions and recommendations for future work. The thesis framework is presented in **Figure 1**.

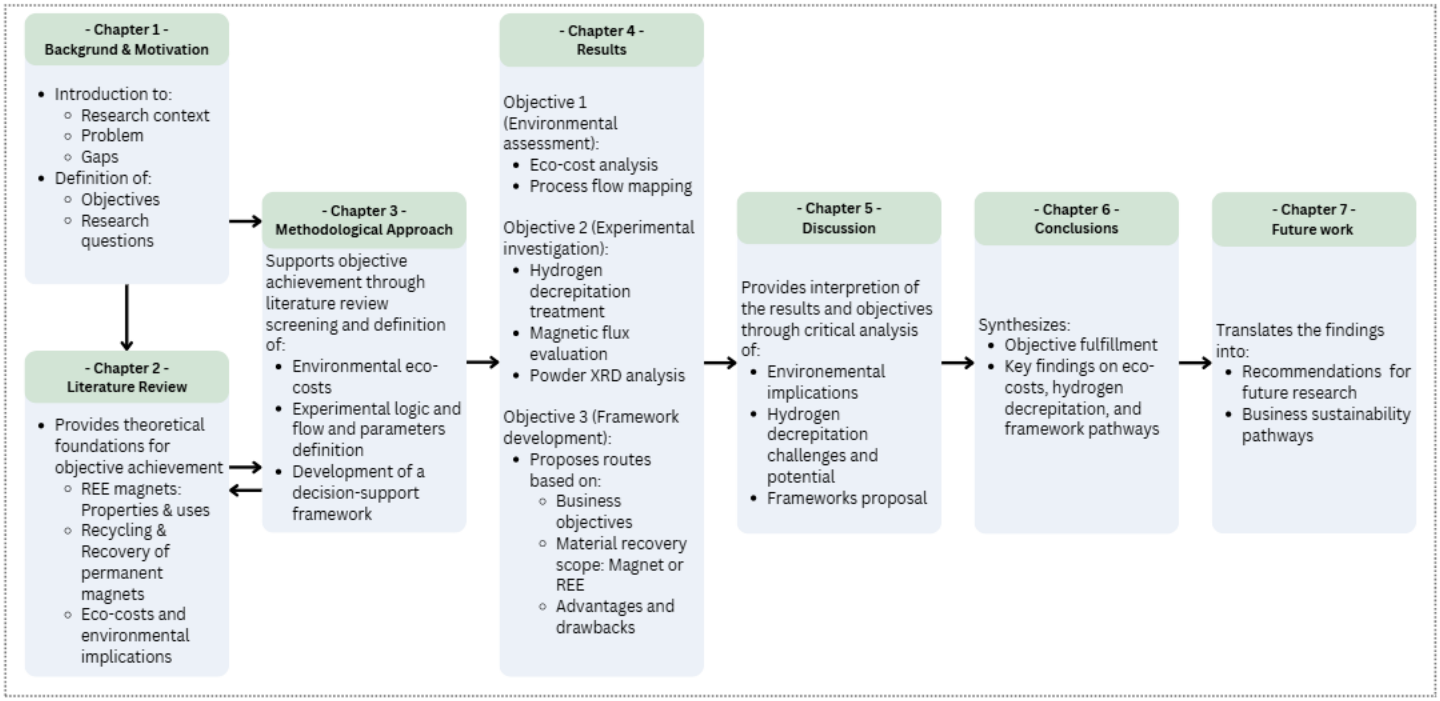


Figure 1: Thesis structural framework.

2 Literature review

2.1 Discovery of magnetite

Magnetic properties have been known since at least 600 BCE, when the Greek philosopher Thales escribed the attraction of objects by magnetic iron ore magnetite. The name magnetite is commonly associated with Magnesia, a region in Thessaly, Greece, where magnetic stones were historically reported. Magnetite pieces, also known as lodestones, were later used in early navigational compasses. Around 1000 CE, studies and practical observations in China showed that lodestones tend to align in a north–south direction, which became fundamental for compass-based navigation. Petrus Peregrinus de Maricourt described the properties of these lodestones, in 1269. Further studies were conducted by Sir William Gilbert in 1600, describing the first systematic magnetism experiments and the magnetic properties of carbon steel [33], [34]. These early observations laid the foundation for the later development of magnetic materials, electric machines, and modern permanent magnet technologies.

2.2 Electromagnetism and its importance

The development of electromagnetism, particularly through the work of Ampère and Maxwell, established the relationship between electric currents, magnetic fields, and mechanical motion. This understanding enabled the development of electric motors and generators during the late nineteenth century, where electromagnetic forces were used to convert electrical energy into mechanical work and vice versa. The first successful EV appeared around 1890, in the U.S.A., and accounted for a significant share of the vehicle fleet by 1900 [34], [35].

However, increased developments for gasoline engines led to a decline in EV popularity. Interest in EV was rekindled around 1970, due to soaring oil prices. Even then, short range capacity and low motor efficiency remained major drawbacks. The performance and travel range of EVs became comparable to gasoline-powered vehicles around the 1990's, coinciding with critical improvements in PMs development [34], [36]. By using the principle of electromagnetic repulsion, the magnetic fields from static electromagnets and permanent magnets repel one another, generating the necessary torque to turn rotors in EVs [23].

2.3 Permanent magnets

Permanent magnets are commonly classified as hard magnetic materials because they can retain magnetization after an external magnetic field is removed. This ability is mainly associated with high coercivity, which allows the magnet to provide a stable magnetic flux during the service life of a device, unless it is exposed to extreme conditions such as high temperature, strong opposing magnetic fields, corrosion, or mechanical damage. Rare-earth permanent magnets, particularly Nd-Fe-B-based magnets, are widely used in applications requiring high magnetic performance, including EV traction motors [37].

Harris [37] has discussed the properties of the rare-earth permanent magnets in extent and detail. Magnetism in ferromagnetic materials can be linked primarily to spin contribution of the electrons in the 3d orbital. While they do not exhibit external magnetic fields when under equilibrium conditions, upon being subjected to an external magnetic field (H), the magnetic domains of the material get aligned in the direction of this applied field, eventually reaching a magnetization saturation (Ms). The magnetization (M) value will decrease in proportion to the quantity of unpaired 3d-electron spins of a material. The value of Ms decreases when temperature increases, up to a point when the magnetic state ceases, which is called the Curie point (Tc). Alternatively, RE metals magnetic behaviour is attributed to the 4f spin-orbit, which presents stronger magneto-crystalline anisotropy. For engineering and material application, magnetic properties of interest are the strength of the H and the net magnetic flux density (B), whose relation can be represented by the formula $B = \mu 0H + J$, where $\mu 0$ is the permeability of free space and J is the magnetic polarization [37].

2.4 Hysteresis and key magnetic properties

Hysteresis describes the lag between the applied magnetic field and the magnetic response of a material. In magnetic materials, this means that the magnetization or magnetic flux density does not immediately follow changes in the applied magnetic field, but depends on the previous magnetic history of the material [38]. The hysteresis loop is therefore an important tool for evaluating permanent magnet performance. Some of the key properties associated with the material's hysteresis characteristics used as main performance indicators are: remanence (B_r) which corresponds to the flux density in the material at zero magnetizing field, in other words, the magnet's ability to retain its magnetization; coercivity (H_c) which corresponds to the magnetizing field strength at which the flux density is nullified, or in other words, it's a measure of the material's resistance to demagnetization, and the maximum energy product $(BH)_{max}$, given by the product of the flux density and the magnetizing field strength at any point in the second quadrant of the hysteresis loop [37].

The use of PMs in modern EV rotors is directly related to its magnetic properties. Torque generation depends on the repulsion between the embedded PMs when subjected to an electric field. That is related to the magnetic field *(B)*. Another essential characteristic is the resistance to demagnetization (H_c) to ensure performance and service life of the rotor. A high performing EV rotor therefore requires PMs that present the maximum magnetic density possible $(BH)_{max}$ to enable compact and lightweight designs, and elevated Curie point (T_c) to guarantee magnetization retention during operation [23]. The hysteresis loop and its properties are shown in **Figure 2**.

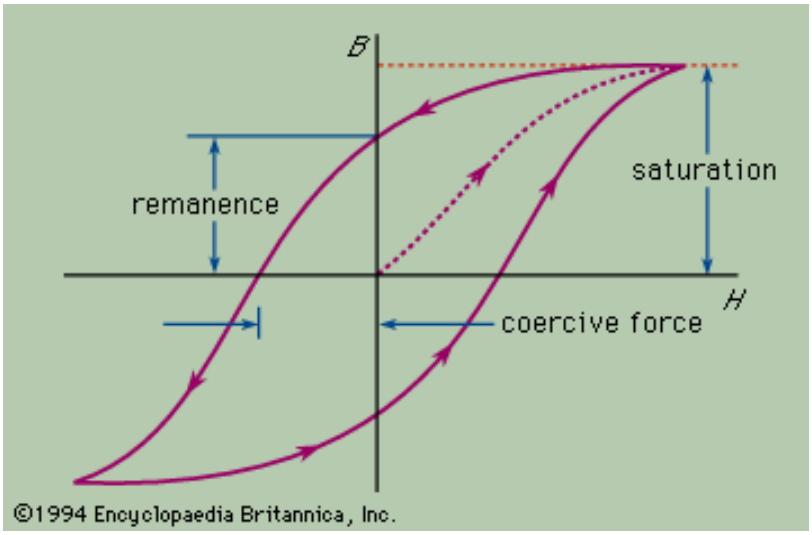


Figure 2: Representation of a magnetic hysteresis loop in ferromagnetic materials. [38]

2.5 Rare-earth permanent magnets

Rare-earth permanent magnets such as $SmCo_5$, Sm_2Co_{17}, and $Nd_2Fe_{14}B$, have represented major breakthroughs in the development of high-performance magnetic materials. $SmCo_5$, first discovered in the 1960's, quickly developed to reach up to about 34 MGOe of maximum energy product $(BH)_{max}$ [39]. In 1982 in Japan, Masato Sagawa and his research team developed the $Nd_2Fe_{14}B$ at Sumitomo Special Metals, which can reach up to a $(BH)_{max}$ of 59 MGOe, revolutionizing the permanent magnet industry [7]. The permanent magnet development history, as well as their comparative volumes and maximum energy products $(BH)_{max}$ can be seen in **Figure 3.**

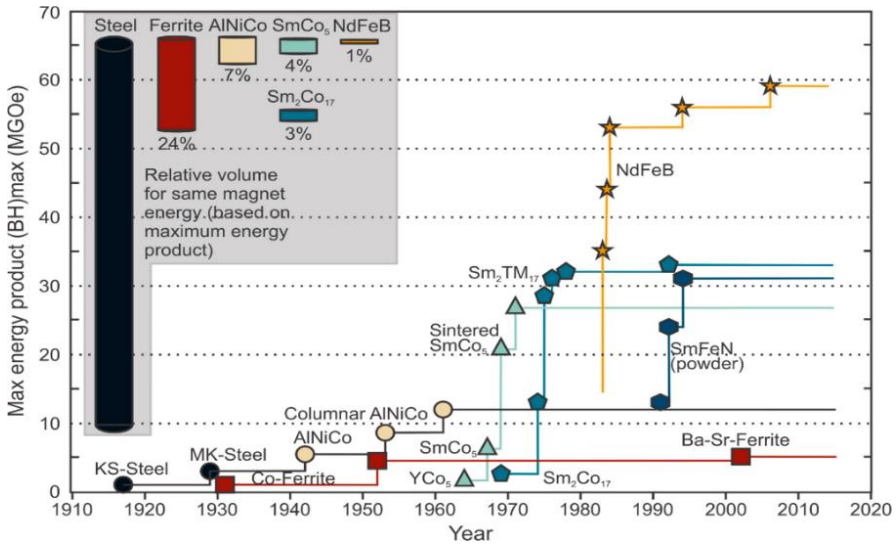


Figure 3: History of permanent magnet development in regard to their corresponding relative volume and maximum energy product $(BH)_{max}$ [7].

Despite their high temperature stability and corrosion resistance, Sm-Co magnets are associated with high production costs, which restrict their use mainly to applications requiring exceptional thermal stability and reliable magnetic performance, such as aerospace, satellites, and medical devices. In contrast, Nd-Fe-B, the strongest permanent magnets, occupy a fourth of the Sm-Co magnet's space for delivering the same maximum energy product, but present a brittle nature, are more susceptible to demagnetization at high temperatures, and quickly oxidize if not properly coated. However, for applications such as EVs, which possess space limitations and demand high performance, Nd-Fe-B magnets are the still considered the best alternatives [7].

Nd-Fe-B magnets have a tetragonal crystal structure, as shown in **Figure 4**. The crystal structure of $Nd_2Fe_{14}B$ is characterized by a specific atomic arrangement in which Nd, Fe, and B occupy distinct crystallographic sites. The Fe atoms form the main magnetic framework, while Nd plays a key role in contributing to the strong magneto-crystalline anisotropy of the material. This crystallographic structure is fundamental to the excellent magnetic properties of Nd-Fe-B magnets [40].

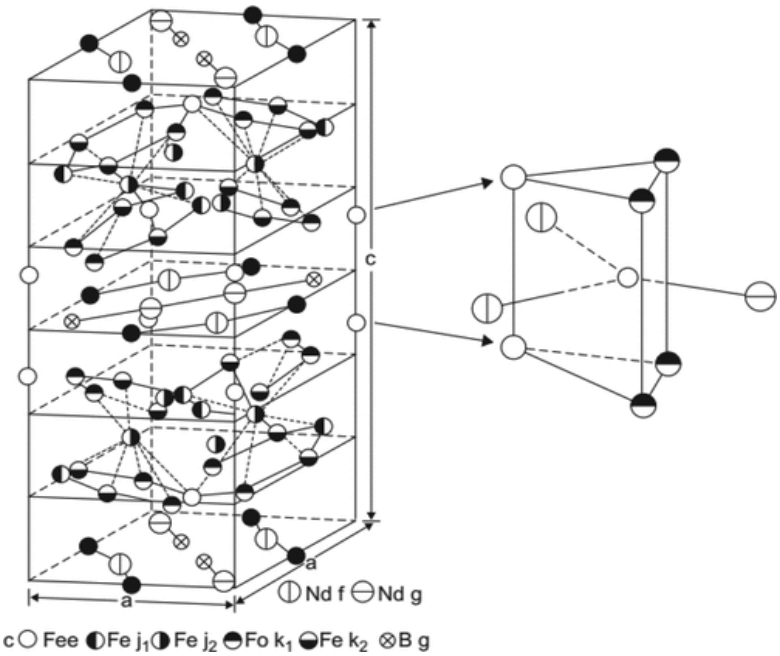


Figure 4: Tetragonal crystal structure of $Nd_2Fe_{14}B$ magnet [40].

However, traditional Nd-Fe-B magnets might not reach the current requirements of high-tech applications for room temperature coercivity or retaining magnetic properties in high temperature environments. The improvement of the thermal resistance of the magnets, and consequent increase in the operational temperature for high-performance applications, can be achieved through the addition of heavy rare earth element (HREE) into the magnet microstructure. Dy/Tb doping has proven especially effective for improving high

temperature performance in applications such as, from around 80°C up to 220°C. However, when introduced through conventional alloying, it can reduce the magnet's Br. Currently, the most used technique for Dy/Tb doping in magnets is grain-boundary diffusion (GBD) due to the coercivity in magnetic materials is controlled at grain-boundary level. When compared to conventional alloying methods, Dy-diffusion into the grain-boundary enhances coercivity properties while minimizing the consequent remanence loss and lowering raw material cost [37], [41], [42]. The Dy diffusion processes vary in efficiency and cost, and the sourcing material can be dysprosium oxide (Dy_2O_3), dysprosium fluoride (DyF_3), dysprosium in the its metal or alloy form, and even dysprosium hydride (DyH_2 or DyH_3), as shown in **Figure 5** [43].

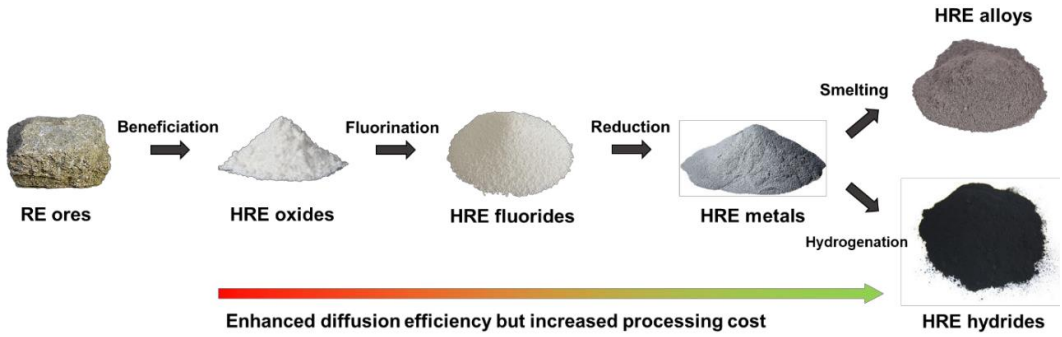


Figure 5: Process cost and efficiency correlation of HRE diffusion sources [43].

Table 1 presents the compositions of Nd-Fe-B permanent magnets for a variety of applications, including highly demanding ones such as EVs and wind turbines. The presence of Dy in the composition of EV magnets is highlighted in comparison to other applications.

Table 1: Permanent magnet composition according to final application. The composition of permanent magnets varies depending on the final application and performance requirements. In particular, EV magnets tend to contain Dy and Pr in addition to Nd, reflecting the need for enhanced coercivity and thermal stability.

Element	**Approximate composition (%)**					
	EV - recycled material [44]	**EV - virgin material** [44]	**Pedelec** [45]	**E-scooter** [45]	**HDD** [46]	**Wind turbine** [47]
Fe	64,57	66,88	66,1	66,4	78,3	67,2
Nd	21,63	18	24	16,6	14	29,6
Dy	3,96	6,15	-	-	0,6	-
Pr	6,43	4,6	6,7	5,4	-	-
B	0,93	1,02	0,9	0,9	6	1
Co	1,74	2,84	1	0,3	-	0,6
Ga	-	0,21	0,2	-	-	
Cu	0,32	0,18	0,2	0,2	-	0,2
Al	0,32	0,12	0,8	0,5	0,7	0,2
Ti	0,1	-	-	-	-	-
Gd	-	-	-	0,7	-	-
Ho	-	-	-	1,8	-	-
Zr	-	-	0,1	0,3	-	-
Ce	-	-	-	6,9	-	-
Nb	-	-	-	-	0,4	-
Tb	-	-	-	-	-	1,2

2.6 Rare-earth raw material production routes for Nd-Fe-B manufacturing

Magnet production involves a series of unit operations, and the sourcing of REEs is critical for the production of high-performance Nd-Fe-B permanent magnets. Rare-earth elements are found in conjunction under the form of rare-earth oxides (REO) in complex rare-earth mineral (REM) ores, such as bastnäsite and monazite, and in ionic adsorption clays (IAC). REE can be classified as light REEs (LREEs) or heavy REEs (HREEs), with Nd belonging to the former, and Dy to the latter. HREEs are scarcer and majorly explored in China, but while LREEs are more abundant than HREEs, it is important to mention that the proportion of Nd in the LREE ores is very low compared to elements such as cerium and lanthanum. As Nd is a key component for high-performing magnets and Dy is a key additive to high temperature PM applications, their availability is of great concern for vehicle electrification [48].

The REO production process will be referenced from China's routes, as this country is currently the prime miner and refiner for REEs in the world. LREE production was obtained from EcoInvent Database Ver. 3.12 and is used as a reference. Sourcing of LREE starts from open pit mining of the REM, which is divided into drilling and blasting operations. A minor quantity of rare-earth ore is obtained from this stage – between 5 - 6 wt.% REO equivalent – and then proceeds to a beneficiation process that is performed in two steps. The first step consists of ball milling and magnetic separation operations, with the aim of isolating the iron ore from the desired REO. During the second step, the remaining ore, which contains bastnäsite and monazite, undergoes a flotation process to concentrate it to 50 wt.% REO equivalent. The obtention of 1 kg of REO concentrate requires approximately 16.5 kg of ore, with obtention of 0.00124 kg and 0.116 kg of Dy and Nd, respectively [49].

The obtention process for neodymium oxide (Nd_2O_3) is based on refining 50 wt. % REO concentrate through a series of unit operations. It is based on technological descriptions from EcoInvent dataset. The concentrate is initially treated with sulfuric acid in an acid roasting stage, followed by leaching in hydrochloric acid for chlorine conversion. The resulting RE-chlorides undergo solvent extraction by addition of organic extractants, and the obtained ions are precipitated in aqueous solutions to produce RE carbonates, which are finally calcinated to yield individual REO [50]. As neodymium must be in its metallic form for permanent magnet manufacturing, it undergoes fused salt electrolysis process, which is considered the habitual industrial reduction process for LREO such as Nd_2O_3 [41].

Although HREE can also be found in ores and are a byproduct of the previously mentioned beneficiation process, China's major HREE production route is sourced from ionic clay deposits located in the country's southern regions and importing from Myanmar [51], [52]. Despite IAC ores presenting lower total REOs content compared to bastnäsite and monazite ores, the percentage of HREEs in the REOs is twenty times higher in the IACs. Furthermore, Dy content in IACs can be forty times the amount from bastnäsite and

monazite. In addition, extracting and processing IACs is comparatively easier, making this route more interesting from an industrial perspective. Sourcing HREE from ionic adsorption clays starts from the drilling of injection wells, into which a leaching solution of ammonia sulphate will be pumped, promoting ion exchange at the clay's surface layer. With the addition of ammonia bicarbonate, rare-earth carbonates are precipitated. This precipitate undergoes an initial hydrochloric acid leaching and the RE carbonates are converted to RE chlorides. A first solvent extraction is performed to separate LREE raffinate, and soda is added to precipitate the HRE carbonates. This is followed by thermal decomposition of the carbonates and a second leaching with hydrochloric acid. The HRE chlorides obtained go through supplementary solvent extraction to separate the elements. Dysprosium chloride is then precipitated via soda addition, and the resulting dysprosium carbonates are calcinated to form dysprosium oxide [53].

2.7 Sintered Nd-Fe-B permanent magnet manufacturing process

The manufacturing process for permanent magnets requires multiple processing stages, starting from Nd, B, electrolytic-grade Fe, and additional alloying elements such as Dy, Pr, Co, Cu, Al, or Ga, depending on the required magnetic performance. During alloying and strip casting process, the desired microstructure of the magnets is established via vacuum induction melting and casting. In this stage metal ingots or powders are melted together. The strip cast flakes produced during the previous stage will then undergo two pulverization processes. At first, the flakes are further broken down into fine friable powder *via* hydrogen decrepitation, followed in sequence by jet milling to even particle size. A powder of mean size distribution of the approximately 5 μm is produced, which is then aligned under a strong magnetic field and compressed in a mold. The compacted material goes through sintering in temperatures above 1000°C. under vacuum or argon atmosphere to reduce oxygen contamination. It is also necessary that temperature is constant to avoid open porosities that can lead to oxidation later in time. The sintered magnets present size reduction and increase in density and may present rough corners and edges that are susceptible to chipping, fracture, and poor coating adhesion. To avoid these, the magnet body needs to be machined through grinding and slicing to attain its desired form and surface. Afterwards, the magnets are cleaned in preparation to undergo Dy-GBD. The diffusion method is based on applying a Dy carrier substance to the magnet's base and submitting it to two heat treatments that do not overcome the sintering temperature. Through the absorption followed by an aging treatment, dysprosium diffuses into the Nd-rich phase of the microstructure, which is located between grains of the magnetic body. The magnet body is electroplated, which can be an optional step, and electrocoated for corrosion protection, Finally, the magnets are magnetized by being exposed to a strong magnetic pulse [40], [41], [42]. An illustration for the general manufacturing process is shown in **Figure 6**.

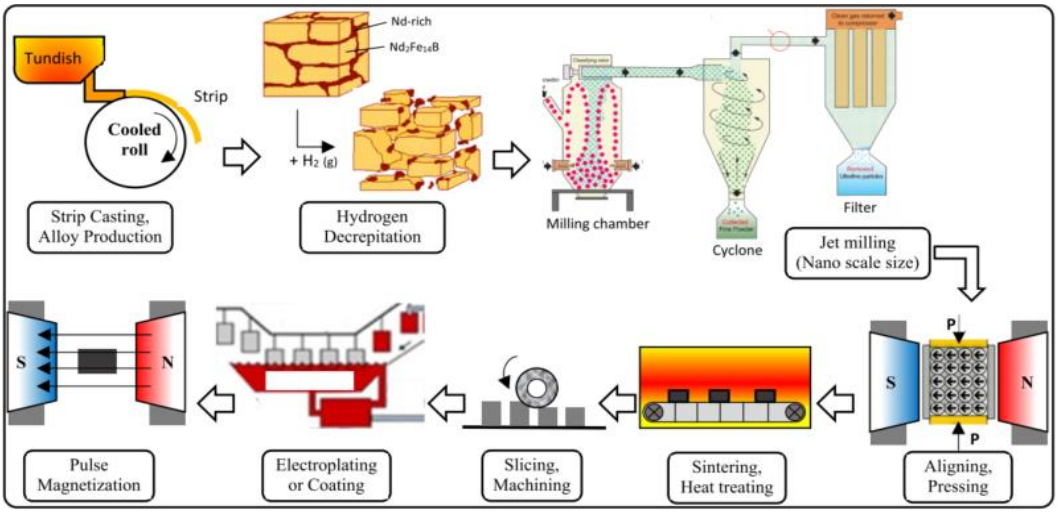


Figure 6: Manufacturing process for sintered NdFeB permanent magnets [40].

2.8 Coating importance and impacts

Nd-Fe-B permanent magnets often contain one or more protective coating layers, which may be metallic, polymeric, or a combination of both. Common coating systems include metallic layers such as Ni-Cu-Ni or Zn, as well as polymeric or epoxy-based coatings. Due to the high chemical reactivity of rare-earth-rich phases in NdFeB magnets, protective coatings are required to improve corrosion resistance, durability, and operational safety. These coatings act as barriers against oxidation and help protect the magnet during service life [26].

Studies have also shown that zinc coatings might alter the NdFeB magnetic properties and Curie temperature due to Zn diffusion into the magnet grain boundary. Hu *et al*. [54] also reported a degradation in magnetic properties due to Zn contamination. They observed that the Zn diffused into the magnet matrix when magnets underwent a low-pressure pack sublimation technique for coating could lead to the formation of Nd-Fe-Zn by consumption of the magnetic phase $Nd_2Fe_{14}B$. While some studies have pointed to improvements in magnetic coercivity and remanence. others have shown that Zn could have detrimental effects on the magnetic properties. The evaluation by Lv *et al*. [55], where 2% in weight of Zn was added to Dy-free NdFeB magnets which were produced through HDDR technology, has shown that coercivity was enhanced and remanence was increased. Horikawa *et al*. [56] discussed the formation of α-Fe by reaction between the diffused Zn and the Fe present in the $Nd_2Fe_{14}B$ matrix of resintered magnets and concluded both Fe and Ni reduce the effective diffusivity of hydrogen in copper alloys.

For nickel inclusion into the magnet matrix, Cui *et al*. [57] found that Ni can enter both the $Nd_2Fe_{14}B$ matrix phase and the Nd-rich phase, affecting grain morphology and magnetic properties. In their study, the optimum coercivity was achieved with 0.3 wt.% Ni as an intergranular additive. However, increasing Ni content led to a decrease in remanence and maximum energy product, $(BH)_{max}$. Therefore, although small, controlled additions of Ni

may influence microstructure beneficially, uncontrolled Ni contamination from coatings may reduce the performance of recycled magnets.

Besides of the effects of coatings on magnet performance, they can also be a challenge during EoL disassembly and recycling. Couto *et al*. [26] have evaluated the effects of coatings in PMs and showed that despite their protective function, coatings can directly influence the recycling process. Their impact is mainly associated with three factors: increased difficulty in accessing the magnet surface, possible inhibition of hydrogen penetration during hydrogen decrepitation, and contamination of the recycled magnet powder by coating-derived elements [30].

De-coating of different coating layers has been evaluated by Couto *et al*. [26] as a key pre-treatment step in PM disassembly. For Zn coatings, ammonia-based solutions were found to successfully dissolve the Zn coating layer, while the same was not possible for Ni coatings due to kinetic constraints. Studies have shown it is possible to dissolve Ni-Cu-Ni coatings by using bromine solutions dissolved in ethanol or dimethylformamide. However, a major drawback of the use of such solutions is that they are highly toxic. A suggestion was the removal of the Ni coating through mechanical separation post hydrogen decrepitation treatment, as the Ni will form flakes that can be removed by sieving the powdered Nd-Fe-B decrepitated matrix. Finally, polymeric coatings may degrade during high-temperature treatments, including thermal demagnetization or adhesive-removal processes. Although this can facilitate coating removal, polymer degradation may generate volatile organic compounds, fumes, residues, or carbonaceous contamination [26]. Therefore, polymeric coatings should not be considered irrelevant to recycling; rather, their thermal stability and degradation behaviour must be considered when selecting pre-treatment and disassembly routes. The extent of these effects depends on the type of coating, its thickness, adhesion to the magnet surface, and the recycling technology used.

Hydrogen embrittlement and hydrogen trapping are well-known phenomena in metallic systems, and alloy composition strongly affects hydrogen uptake, diffusion, and trapping behaviour. In copper-based alloys, Ni has been reported to increase hydrogen trapping capacity and improve resistance to hydrogen-driven embrittlement [58], [59], [60], [61]. However, there is limited research on the H_2 diffusing behaviour in Ni-Cu-Ni alloy, especially if used for magnet coating applications. Therefore, several important questions in Nd–Fe–B magnet recycling are needed to be answered: what type of coating Nd–Fe–B magnets have, whether a Ni–Cu–Ni coating acts as a barrier that prevents hydrogen from accessing the reactive magnet phases. This is important because hydrogen decrepitation requires hydrogen to pass the coating layer and reach the Nd-rich grain-boundary phase and the $Nd_2Fe_{14}B$ matrix. If the metallic coating acts as a barrier to hydrogen diffusion, the decrepitation reaction may be delayed, incomplete, or fully inhibited.

Therefore, to evaluate the hydrogen decrepitation potential of intact alloy-coated permanent magnets, the presence, type, thickness, and integrity of the coating must be considered. For EV rotor magnets, this issue is even more critical because the magnet

surface may also be covered by adhesive bonding layers, and present limited exposed surface area inside rotor pockets. Understanding how these protective barriers influence hydrogen access is therefore essential for developing efficient disassembly routes and assessing the circularity potential of EoL permanent magnets.

2.9 Disassembly and recycling techniques

Recycling critical raw materials is currently one of the major industrial challenges associated with resource depletion, supply-chain vulnerability, and the increasing demand for clean-energy technologies. This challenge is particularly relevant for permanent magnets, since high-performance Nd-Fe-B magnets depend on rare-earth elements that are both strategically important and geographically concentrated in supply. According to Regulation (EU) 2024/1252, the materials listed in **Table 2** are considered critical raw materials for the EU. Among these, light rare-earth elements (LREEs), heavy rare-earth elements (HREEs), boron, cobalt, copper, aluminium, and nickel are particularly relevant to EV powertrains and permanent magnet technologies. More than 23% of the global supply of REOs is currently directed to the manufacturing of Nd-Fe-B permanent magnets, and accounts for over 60% of the market value of RE-based materials [14], [62].

Despite being one of the major producers of high-performance PMs the European Union strongly depends on critical raw material imports. Currently, REE recycling amounts to less than 1% globally, which can be attributed to technological barriers, poor management of waste, and lack of governmental backing [30], [63]. In the EU, this situation is now bound to change due to new regulatory measures. With the implementation of the CRMs Act in 2024, the EU must guarantee that at least 25% of its critical raw materials come from secondary sources, in other words, from recycling processes, and that directly implicates evaluating the recycling and recovery of RE permanent magnets [11], [13], [30].

Disassembly and recycling processes have been evaluated in various EoL streams targeting CRM recovery, including EVs, and consumer electronics [24], [27], [28], [64]. The disassembly of PM from EV synchronous rotors has been evaluated from manual to automated robotic perspectives [27], [65]. The findings of those studies have highlighted a key disassembly challenge that PMs are often strongly bonded with adhesives, making separation difficult. They also indicate that more data and development are needed to be done to adequately perform efficiently, non-destructive, automated disassembly of the rotor and their embedded PMs. The general structure of a synchronous rotor and the location of the embedded magnets is presented in **Figure 7** to allow for better visualization of the disassembly challenges.

Table 2: Critical raw materials for the EU according to Regulation (EU) 2024/1252. Materials directly relevant to permanent magnet production or EV powertrain systems are highlighted in bold. [14]

Antimony	Arsenic
Bauxite / Alumina / Aluminium	Baryte
Beryllium	Bismuth
Boron	**Cobalt**
Cooking coal	**Copper**
Feldspar	Fluorspar
Gallium	Germanium
Hafnium	Helium
HREE	**LREE**
Lithium	Magnesium
Manganese	Graphite
Nickel (battery grade)	Niobium
Phosphate rock	Phosphorus
Platinum group metals	Scandium
Silicon (metal)	Strontium
Tantalum	Titanium (metal)
Tungsten	Vanadium

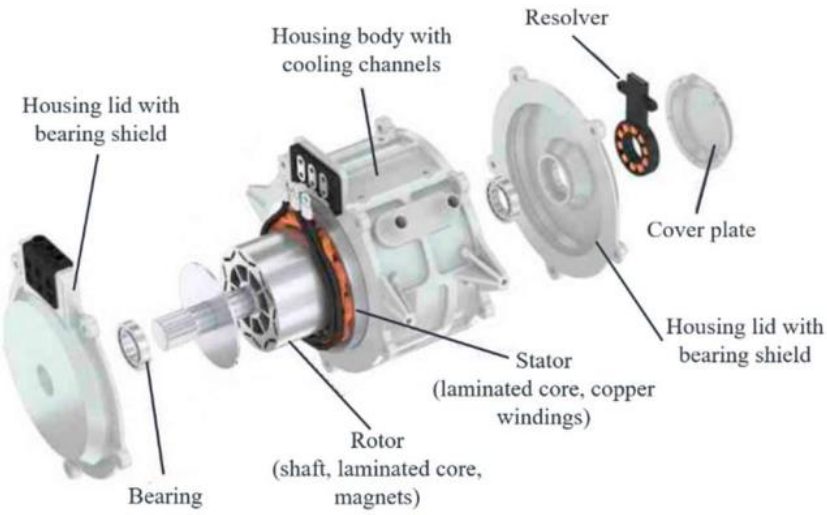


Figure 7: Main components of a permanent magnet synchronous rotor [27].

Considering the safety requirements for magnets embedded in rotors that operate under high rotation speeds and elevated temperatures, it is critical that these PMs are strongly fixated to the rotor's electric steel structure. The adhesive bonding can be explained as a combination of mechanisms that include chemical cross-linking, mechanical anchoring, dipole-dipole interactions, hydrogen bridging bonds, among others. Due to this complexity, removing the magnets from the rotor pockets is not a simple task to be achieved in a pure mechanical, non-destructive manner [25], [66]. The need for preliminary thermal treatments for demagnetization and fragilization of the adhesive materials fixating the magnets, besides performing appropriate post treatment of the magnets is also discussed [25]. In their study, Veller *et al*. [25] have pointed out that using solely mechanical separation is not sufficient to achieve complete separation, heating the stacks above the glass transition temperature (T_g) together with the application of mechanical shearing can yield results even at low forces, and that thermal treatments above Curie temperature allow for easier PM detachment from the structure. They additionally point out that modifying the adhesive with expandable particles and combining it with heat treatment and electrical voltage application also leads to disassembly. However, it is important to point out that thermal treatment of adhesive substrates can present environmental impacts connected to the release of volatile organic compounds (VOCs) and hazardous fumes, due to the degradation of the polymeric substances that form the adhesive [66].

Besides disassembly, ongoing research on REE recovery from magnets includes several recycling routes. These are commonly grouped into direct reuse or magnet-to-magnet recycling, waste-to-alloy recycling, and waste-to-REE recovery [63]. Permanent magnets can come from diverse applications, each presenting its own distinct compositions as presented previously in **Table 1**, and these compositions are linked to the magnetic properties desired of these PMs, as is related in **Table 3**. Mixing these compositions will inevitably lead to change in properties, hindering re-sintering without previous pre-

treatments and composition adjustments [67]. Therefore, the recycling of Nd-Fe-B permanent magnets involves several challenges, including the heterogeneous composition of magnets in mixed waste streams, the inherent brittleness of the material, and rapid microstructural oxidation after protective coatings are removed.

Table 3: Performance parameters of permanent magnets related to application.

Performance parameters	**EV - recycled material [44]**	**EV - virgin material [44]**	**Pedelec [45]**	**E-scooter [45]**	**HDD [46]**	**Wind turbine [47]**
Br (T)	1.3	1.2	1.33	1.25	1.315 ± 0.05	1.4
bH_c (kOe)	12.6	11.5	-	-	-	-
iH_c (kOe)	>20	19	-	-	12.06 ± 0.63	16.92
H_{cj} (kA.m^{-1})	-	-	1334	1248	-	-
(BH)max (MGOe)	40.7	34	43.23	37.07	40.84	47.52
Operating Temperature (°C)	180	180	-	-	-	-

The Nd-Fe-B recycling techniques can be divided into long-loop and short-loop recycling processes, where the first refers to the recovery of the elements of interest in a waste-to-REE approach, and the latter focuses on minimizing the process steps to reprocess magnets, in other words, waste-to-alloy or waste-to-magnet approaches [29], [62]. Current short-loop processes are the calcio thermic reduction diffusion (CTRD), and the hydrogen assisted techniques – such as hydrogen decrepitation and HDDR process [26], [29], [62], [63]. The former – CTRD – is currently a recycling alternative to the Nd-Fe-B sludge generated during the manufacturing process. It is an efficient and environmentally sustainable way of recovering Nd-Fe-B particles that are oxidized and contaminated by ater and oil [62]. The HDDR process will be further discussed in section 2.9.2 of this report.

Long-loop recycling is based on more traditional approaches such as hydrometallurgy and pyrometallurgy. The main recycling methods will be explained in sequence, with focus on the hydrogen assisted methods as the purpose of this study, and a schematic representation of these main processes is presented in **Figure 8**.

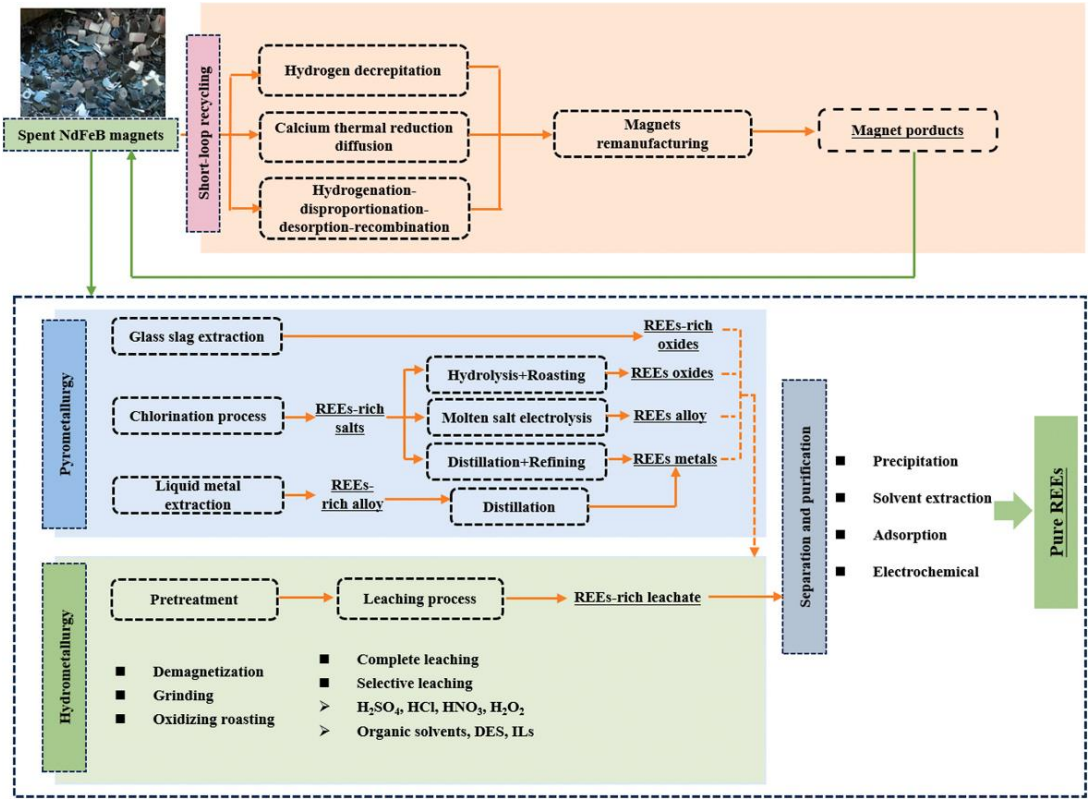


Figure 8: Schematic diagram of recycling routes for waste NdFeB magnets [62].

2.9.1 Long-loop recycling

Long-loop recycling routes recover valuable elements or compounds from Nd-Fe-B magnet waste by breaking the magnet material down into chemical intermediates. These routes are often referred to as waste-to-REE approaches because the final aim is usually the recovery of rare-earth elements or rare-earth compounds rather than the direct reuse of the magnet alloy. The most common long-loop approaches are hydrometallurgical and pyrometallurgical processes.

Hydrometallurgical routes recover valuable metals from Nd-Fe-B magnets primarily through leaching, in which the target elements are dissolved into an aqueous phase. These techniques are industrially mature, and are largely adopted due to high leaching efficiency, lower energy consumption in comparison to pyrometallurgical routes and the use of well-established systems for a variety of applications. Following appropriate pre-treatment – most commonly demagnetization – the leaching kinetics of Nd-Fe-B magnets can be significantly enhanced [62]. Several leaching approaches have been reported, including complete leaching with or without roasting, as well as selective leaching [62], [63].

A key advantage of hydrometallurgical processing is its flexibility: depending on the flowsheet, it can be tailored to recover either individual metals or mixed rare-earth element (REE) streams. Moreover, these methods are generally applicable to complex and heterogeneous magnet waste streams, including different magnet types and grades. In

practice, hydrometallurgical processes are valued for their controllability, operational predictability, and high material yields. However, these benefits come with notable drawbacks. It requires large amounts of water and reagent chemicals, which also lead to high generation of sludge and wastewater that demand additional treatment. Among emerging variants, ionic-liquid-based extraction has been proposed as a novel approach for REE separation. Ionic liquids are often presented as potentially more selective or environmentally favourable alternatives to conventional solvent systems [68]. However, most current applications remain at laboratory or proof-of-concept scale, and broader technological readiness for industrial-scale deployment has not yet been demonstrated [29], [30], [32], [62], [63].

Pyrometallurgical routes, despite not requiring large water and reagent consumption, are very energy-intensive due to high temperature processing. The elevated temperatures are used to alter the chemical state of the elements, making separation and recovery easier. The pyrometallurgical processes can be divided into three different types, according to the mechanism used to recover the REEs, which can be: (i) selective oxidation reactions, (ii) salt-roasting processes, and (iii) liquid metal extraction. These routes may produce rare-earth-containing alloys or mixed compounds, which usually require additional purification before individual high-purity REEs can be obtained [19], [30], [62], [63]. Downstream refining is commonly required, for instance, via hydrometallurgical treatment or molten-salt electrolysis, to produce high-purity rare-earth products [62], [63].

Key advantages of these routes include their suitability for remelting magnet alloys and recovering transition metals from the magnet matrix. They are generally less complex than hydrometallurgical flowsheets and typically require substantially lower water consumption. However, they also have drawbacks include higher energy consumption, lower material yield compared to hydrometallurgy, and the routes are often not economical for small feed quantities [62], [63]. As presented in **Figure 9**, the techniques from hydrometallurgical and pyrometallurgical routes require operations to recover the REE from the spent magnets. These are associated with the previously mentioned drawbacks and higher environmental impacts, in contrast to magnet-to-magnet recycling routes focused on hydrogen techniques [69], which will be presented in sequence.

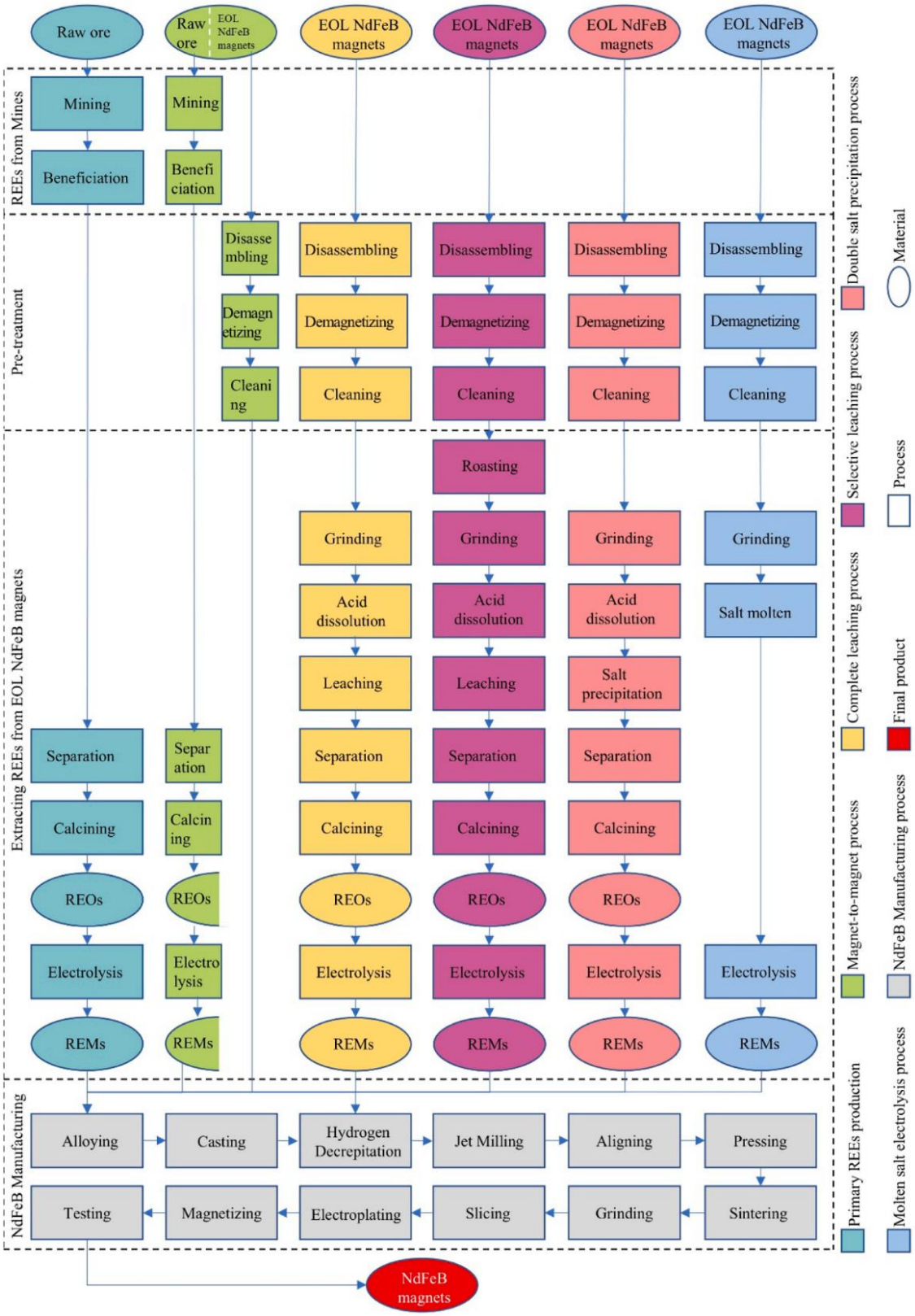


Figure 9: Process comparison between manufacturing and recycling of permanent magnets [69].

2.9.2 Short-loop recycling

Hydrogen, as the smallest known atom, easily penetrates the grain boundary structure of a variety of materials, and presents high reactivity. The basis for the Hydrogen decrepitation (HD) process is the characteristic mechanism of material failure through brittle fracture, which occurs by hydrogen diffusion into the material's grain boundary and the creation of pressure points in the weakest spots of the microstructure. This internal pressure leads to the spread of microcracks in the structure and decrease of the fracture strength, ultimately causing the material to break apart. HD is used as step in powder metallurgy that is used in the manufacturing of sintered magnets. The inclusion of this step considerably reduces the processing cost, as it can reduce the material to fine powder at room temperature and pressure of 1 bar, and allows the process time in the milling step to be significantly reduced [70], [71].

HD also have been proposed to be used for direct recycling by exposure of sintered Nd-Fe-B magnets to hydrogen in order to break them down into a demagnetized powder to be used in the production of new magnets, bonded or sintered [30], [72], [73], [74]. Zheng et al. [62] reported that the use of HD can additionally help with safer and more efficient recovery methods by eliminating the need for manual dismantling. This powder, with particle size usually ranging from 6 - 600 μm – which can vary depending on process parameters. It contains $Nd_2Fe_{14}BH_X$ hydride and can subsequently be used in the production of new magnets [30], [40].

One of the preliminary processes discussed in literature for treating EoL Nd-Fe-B magnets from EVs with the HD method is the disassembly and separation of magnets from the rotor [26], [29], [30]. It also claimed that coatings, such as the Ni-Cu-Ni or Zinc metallic alloys, could possibly hider the penetration of the hydrogen atoms into the matrix [26], [29], [62], [75] , should be removed prior to HD treatment to maximize the surface area available for hydrogen penetration [26], [29], [30], [40], [62], [67], [75]. Michalski et al. [76] reported HD and HDDR treatments of standard Nd-Fe-B magnets with protective nickel coating from scrap hard drive disks. The used magnets presented coating damage from dismantling, but the authors have not applied any extra coating removal intervention before the HD and HDDR treatments. Their results show that the magnet underwent decrepitation, and the nickel coating formed large flakes that could be removed through sieving, further demonstrating the effectiveness of hydrogen decrepitation techniques.

Hydrogen assisted techniques have been considered one of the most sustainable alternatives to traditional hydrometallurgical and pyrometallurgical routes. Besides much lower demand for energy and chemical reagents, hydrogen decrepitation does not generate waste and has much lower emissions, being more environmentally friendly [63]. However, it is important to highlight that hydrogen assisted recycling processes are not suitable for mixed or shredded material, as it will increase the contamination levels of the resulting powder, leading to a decrease in magnetic properties [29], [67].

To produce new high-performance sintered magnets from recycled EoL magnets, different technologies combinations from the recycling routes described above can be used, as shown in **Figure 9**. These include complete leaching, selective leaching, double-salt precipitation, molten-salt electrolysis, and magnet-to-magnet recycling routes [69]. The choice of recycling route depends strongly on feedstock quality, coating condition, contamination level, magnet composition, and whether the target is REE recovery, alloy recovery, or direct magnet remanufacturing.

2.9.2.1 Mechanism

2.9.2.1.1. Hydrogen Decrepitation

HD occurs through the reaction of hydrogen with the hydrogen-sensitive phases in Nd-Fe-B magnets, mainly the Nd-rich grain-boundary phase and, subsequently, the $Nd_2Fe_{14}B$ matrix phase. The process can occur at relatively low temperatures, typically within the range of 25–400°C, depending on magnet composition, surface condition, hydrogen pressure, and coating integrity. For hydrogen pressures below 1 MPa, the HD process has been described as proceeding through four main stages: surface activation, slow hydrogenation of the Nd-rich grain-boundary phase, rapid hydrogenation of the $Nd_2Fe_{14}B$ matrix phase, and slower hydrogenation of the inner magnet bulk [77].

Initially surface activation stage for coated magnets occurs, followed by the diffusion of H_2 into the magnet and expansion of the lattice volume until the structure cracks. Hydrogen reacts exothermically with the Nd-rich phase at approximately room temperature. The released heat allows a second stage to take place, where hydrogen is then absorbed by the $Nd_2Fe_{14}B$ matrix phase. The process is completed when the Nd-rich and $Nd_2Fe_{14}B$ matrix phases are fully hydrogenated and H_2 absorption ceases. At these process parameters, the HD process leads to the formation of two phases: $NdH_{2.7}$ and $Nd_2Fe_{14}BH_{2.7}$.

The process can be expressed by the reactions presented in Equations (1) and (2), with corresponding lattice volume expansions of $\Delta V1 = 16\%$ and $\Delta V2 = 5\%$ [40], [76]:

Grain boundary reaction: $Nd + H_2 \rightarrow NdH_2$ Equation (1)

Grain reaction: $Nd_2Fe_{14}B + H_2 \rightarrow Nd_2Fe_{14}BH_2$ Equation (2)

Lixandru et al. [67] carried out the hydrogen decrepitation of magnet samples at room temperature, while maintaining the process pressure under 100 kPa. Li et al. [77] has evaluated the effects of H_2 pressure in the decrepitation of the magnets, and concluded that hydrogenation process accelerates with increasing pressure, and surface activation accelerates with increasing pressure up until 6 MPa, decreasing reaction times. However, excessive pressure increase can hinder pulverization capacity of the HD process, presented in the investigation by Habibzadeh & Gökelma on the formation of a dense NdH_2 surface layer [75].

Michalski et al. [76] evaluated that temperatures above 170°C are necessary to successfully decrepitate NdFeB magnets due to the presence of nickel coating. They evaluated the HD process in the temperatures 50°C, 100°C, 200°C, and 300°C, at both 50 kPa and 200 kPa of hydrogen pressure. It was followed by dehydrogenation in a temperature range of 720 - 820°C. The study concluded that the best magnetic properties occur at the temperature of 50°C for hydrogenation with a process pressure of 50 kPa. However, at lower pressures initialization time is longer, so a hybrid process was suggested by quickly applying a 200 kPa pressure initially. This pressure combination substantially improves process efficiency by reducing completion time. For dehydrogenation, they recommend the temperature of 780°C.

Establishing the process parameters for effectively conducting the HD and the Hydrogenation-Disproportionation-Desorption-Recombination (HDDR) treatment of EoL permanent magnets is a key step. Michalski et al. [76] reported that the HD process accelerates with temperature and pressure increase. In this study they propose a hybrid process based on the combination of different pressures with the desired temperature to minimize processing time and maximize efficiency.

According to Habibzadeh et al. [40] the hydrogen decrepitation process is suitable for low temperatures, with usual range of 25°C to 400°C, where H_2 pressure can be varied from 0.5 up to 15000 kPa. The temperature parameters used for dehydrogenation are in the range of 600°C to 1050°C. Their recommended settings to avoid undesired grain growth is to conduct hydrogenation at 100°C and 100 kPa, and to keep dehydrogenation to temperatures under 700°C.

Habibzadeh & Gökelma [75] investigated the pulverization mechanism of NdFeB magnets during hydrogen decrepitation at temperatures between 25 - 400 °C and hydrogen pressure of 1 bar. They examined the effects of NdH_x and $Nd_2Fe_{14}BH_x$ formation when H_2 atoms penetrate the Nd-rich grain boundary phase and interstitial $Nd_2Fe_{14}B$ matrix grains sites. The experiments revealed that during the heating the Nd-rich and $Nd_2Fe_{14}B$ matrix phases are converted to Nd_2O_3 and NdH_2. and four different types of $Nd_2Fe_{14}BH_x$. Their proposed mechanism for H_2 diffusion is presented in **Figure 10**.

They discuss the precipitation of NdH_2 in the outer layers of the Nd-rich phase. Their study showed that, at temperatures below 300°C, the formation of NdH_2 at the surface creates a barrier that hinders further diffusion of H_2 atoms into the bulk of the magnet. According to their findings, the conversion of the Nd-rich phase into neodymium hydride begins at approximately 150°C and becomes complete near 288°C, as the diffusivity of hydrogen atoms increases with temperature, allowing them to penetrate the NdH_2 layer. However, significant volume expansion and efficient pulverization depend primarily on the hydrogenation of the $Nd_2Fe_{14}B$ matrix grains rather than the hydrogenation of Nd-rich phase due to the much higher composition percentage of the former. The largest magnet volume expansion was identified around 100°C [75]. However, at temperatures below 300°C, although the reaction is thermodynamically favourable, kinetic factors are

prevalent, and the reaction time is prolonged, where the complete reaction of the Nd-rich layer would require approximately 18 hours at 236°C while it should take 2 hours at 303°C [75].

As discussed by Habibzadeh & Gökelma [75], excessively high hydrogenation temperatures may not be optimal for pulverization. Their best pulverization results were obtained at 100–200 °C, where the highest content of $Nd_2Fe_{14}BH_{4.73}$ and the finest particle yield were observed. Their hydrogenation treatment and subsequent argon flushing were conducted at 1 bar. In their investigations, they conducted hydrogen decrepitation for two hours and proceeded with another two hours for argon flushing and cooling of the magnet samples.

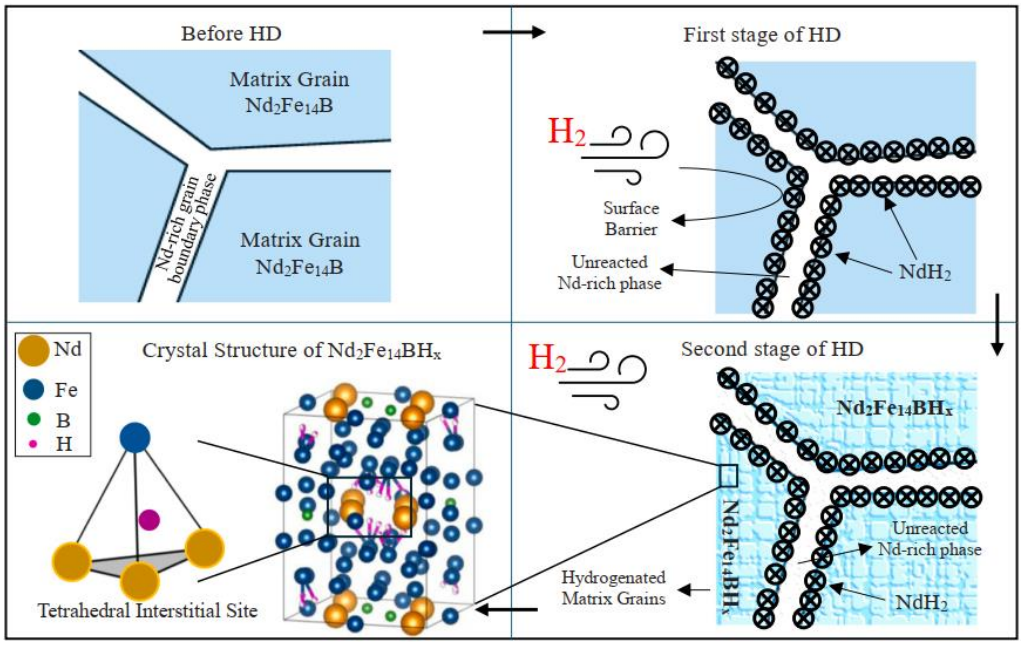


Figure 10: Schematic modelling of hydrogen diffusion into Nd-rich grain boundary and matrix grains in NdFeB magnets [75].

2.9.2.1.2. Hydrogen, Disproportionation, Desorption, and Recombination (HDDR)

Hydrogenation, disproportionation, desorption, and recombination (HDDR) process is another hydrogen-assisted route used to produce magnet powder with useful coercive properties. Unlike simple hydrogen decrepitation, which mainly fragments the magnet into hydrogenated powder, HDDR involves a controlled phase transformation of the $Nd_2Fe_{14}B$ phase at elevated temperature. In the HDDR process, the final powder can develop a very fine microstructure due to the decomposition and subsequent recombination of the $Nd_2Fe_{14}B$ phase. The process can produce fine $Nd_2Fe_{14}B$ grains surrounded by a thin Nd-rich grain-boundary phase, which contributes to improved coercivity, H_{cj}, and maximum energy product, $(BH)_{max}$, in the final magnet [40], [78].

The HDDR process requires elevated temperatures, where the magnetic phase $Nd_2Fe_{14}B$ is decomposed into NdH_2, Fe, and Fe_2B by submitting the material to thermal treatment from 750-950°C under hydrogen atmosphere. It is then followed by desorption of the hydrogen by heating under vacuum at 800°C, recombining the NdH_2, Fe, and Fe_2B into fine $Nd_2Fe_{14}B$ grains with particle size smaller than 1μ [40], [70], [78].

The HDDR reaction process at each stage can be described by equations 3, 4, 5 and 6 [40], and shown in **Figure 11**. The full process reaction can be represented as shown in equation 7 [79]:

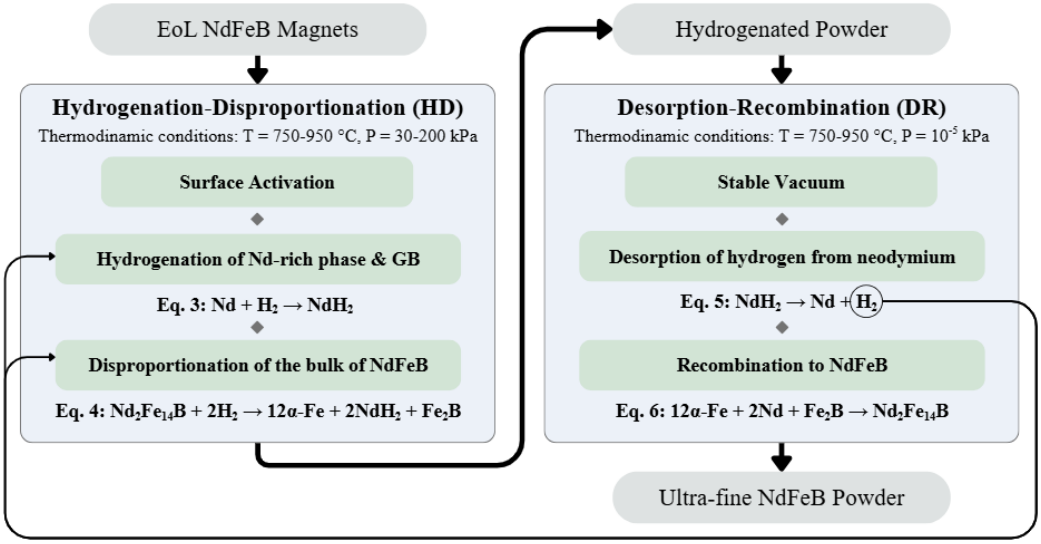


Figure 11: Mechanism flow for recycling an EoL Nd-Fe-B magnet through the HDDR process. Adapted from *[40]*.

Hydrogenation of Nd-rich phase and GB: $Nd + H_2 \rightarrow NdH_2$ Equation (3)

Disproportionation of NdFeB: $Nd_2Fe_{14}B + 2H_2 \rightarrow 12\alpha\text{-}Fe + 2NdH_2 + Fe_2B$ Equation (4)

Desorption of H_2 from Nd: $NdH_2 \rightarrow Nd + H_2$ Equation (5)

Recombination to NdFeB: $12\alpha\text{-}Fe + 2Nd + Fe_2B \rightarrow Nd_2Fe_{14}B$ Equation (6)

HDDR process: $Nd_2Fe_{14}B + (2\pm x)H_2 \leftrightharpoons 2NdH_{(2\pm x)} + 12Fe + Fe_2B + \Delta H$ Equation (7)

Gutfleisch and Harris [79] have discussed the HDDR process variables and their effects. For instance, the final powder's magnetic properties are affected by the initial state of the alloy, hydrogen pressure, hold temperature/time/pressure parameters, evacuation temperature, evacuation time, ultimate vacuum pressure, cooling rate, and further treatments applied to the powder. They concluded, among others, that the production of anisotropic powder is especially sensitive to the parameters applied in the disproportionation stage, in which raising temperature to 900°C leads to improved properties. HDDR process parameters have also been discussed by various authors in pursuit of optimal conditions and better yield results.

Lixandru et al. [67], conducted the HDDR process' hydrogenation-disproportionation in a pressure range from 30 - 100 kPa, reducing it to 1 kPa for the desorption and recombination step. Temperature varied from 780 - 840 °C during disproportionation and held at the latter value until desorption and recombination were concluded, at which point the sample was cooled to room temperature. Gutfleisch and Harris [79] held the sample in disproportionated state for two hours. Lixandru et al. [67] conducted the HDDR process' hydrogenation-disproportionation in an approximate total time of three hours, where both the (HD) and (DR) steps were performed for one hour and half. **Table 4** presents a summary of the process parameter ranges for hydrogen-assisted methodologies, as reported by literature.

According to Habibzadeh et al. [40] the HDDR process' suitable parameters are temperatures in the range of 750°C to 950°C, with pressure varying from 10^{-5} kPa, in other words vacuum, to 100 kPa depending on the process stage. During the hydrogenation-disproportionation (HD) stage, pressure could vary from 30 - 200 kPa and then be reduced to 10^{-5} kPa for, or after, the desorption-recombination (DR) step depending on the preferred process dynamic. They also recommend keeping the disproportionation process temperature below 900°C and maintaining pressures under 100 kPa, because increasing the pressure beyond this threshold led to a deterioration in magnetic properties.

Table 4: Summary of reported hydrogen-assisted processing parameters for NdFeB magnets.

Process	Temperature (°C)	Pressure	Main purpose	Key limitation	Ref.
HD	25°C to 400°C	50 – 200kPa	Magnet pulverization	Sensitive to surface availability and contamination	[40], [76]
Dehydrogenation	600°C to 1050°C	Vacuum	Hydrogen removal from powder	Risk of grain growth and oxidation if poorly controlled	[40], [76]
HDDR – (HD)	750-950°C	30 - 200 kPa	Magnet pulverization	Requires strict process control and high temperatures	[40], [67], [79]
HDDR – (DR)	750-950°C	Vacuum - 200 kPa	Formation of fine coercive powder	Requires strict process control and high temperatures	[40], [67], [76]

2.9.2.2 Limitations

Although hydrogen-assisted recycling is often considered more resource-efficient and less chemically intensive than hydrometallurgical and pyrometallurgical routes, efficient implementation requires careful control of several material and process-related factors. The first major limitation is hydrogen access to the magnet surface. For hydrogen decrepitation to occur, hydrogen must reach the Nd-rich grain-boundary phase and the $Nd_2Fe_{14}B$ matrix. Therefore, the hydrogen permeability and integrity of protective coatings must be understood. In many cases, metallic or polymeric coatings may need to be partially or fully removed before hydrogen treatment to enable efficient decrepitation and to prevent coating-derived residues from negatively affecting the magnetic properties of the recycled material [30], [40], [62], [79].

A second limitation is the strong sensitivity of hydrogen-assisted processes to processing parameters. Time, hydrogen pressure, temperature, feedstock composition, sample geometry, surface condition, and batch size can all influence hydrogen absorption, reaction kinetics, particle-size distribution, and final powder quality. Insufficient temperature or pressure may result in incomplete decrepitation, whereas excessive temperature or pressure may reduce pulverization efficiency, promote grain growth during post-treatment, or increase oxidation risks [30], [40], [62], [79]. Therefore, careful parameter optimization is required to obtain high-quality recycled powder.

Feedstock variability is another major challenge. Achieving compositional homogeneity is difficult when magnets originate from different waste streams, since even magnets from similar applications can contain different amounts of Nd, Pr, Dy, Tb, Co, Cu, Al, Ga, or other alloying elements. This compositional variation can affect hydrogen reaction behaviour, powder quality, and the magnetic properties of recycled products. Mixed or poorly sorted magnet waste may therefore be unsuitable for direct magnet-to-magnet recycling without additional composition adjustment or refining. Oxidation is also a critical limitation. Nd-Fe-B magnets are highly sensitive to air and moisture, especially after coating damage or powder formation. Oxidation can degrade magnetic properties, alter the phase composition, and reduce the efficiency of hydrogen decrepitation [26], [30].

Maintaining an inert atmosphere during handling, treatment, cooling, and post-processing is therefore essential for preserving powder quality. However, this requirement adds complexity, especially in industrial-scale recycling where magnets may be mechanically removed, damaged, or exposed to air before processing. For EV rotor magnets, these limitations are even more significant. The available surface area is often restricted because the magnets are embedded inside rotor pockets and bonded to the electrical steel structure by adhesives. Adhesive layers, polymeric coatings, and metallic protective coatings may all reduce hydrogen access, increase reaction time, or completely inhibit decrepitation. Furthermore, mechanically disassembling magnets from rotor stacks without damaging the coating or magnet surface may be difficult at industrial scale. If pre-treatments are required to remove coatings or adhesives, they must be designed carefully to avoid oxidation, contamination, excessive energy use, or harmful emissions from polymer degradation.

Another practical challenge is the limited availability of detailed information about EoL magnets. Since magnets are often covered by external corrosion-protection coatings, visual identification of coating type, magnet grade, or damage state may not be possible. In addition, recyclers may not have access to complete information about magnet composition, coating structure, thermal history, or original manufacturer specifications. This lack of material traceability complicates the design of standardized hydrogen-assisted recycling processes for EV permanent magnets.

2.10 Environmental eco-costs and impacts, recycling importance, and challenges

As previously presented, the manufacturing of new, virgin Nd-Fe-B permanent magnets is dependent on the extraction of REE from either ores (LREE) or ionic adsorption clays (HREE). The sourcing of these materials is based on a series of unit operations comprised of hydro- or pyro- metallurgical processes, which have high associated environmental impacts.

Environmental databases such as Ecoinvent and IDEMAT 2026 were used to gather foreground and background data respectively, for the manufacturing process of new Nd-Fe-B permanent magnets [80], [81]. Eco-costs are an economic measure of the environmental burden associated with a given activity. In other words, it can be said eco-costs represent the necessary investments to reduce the impacts related to that activity.

Upon evaluation of the datasets, it was identified that downstream and upstream impacts of Nd-Fe-B magnet production presented environmental costs of € 83.30 per new 50 MGO magnet, and € 109.13 for the corrosion resistant 40 MGO grade. Additionally, related impacts to the production of 1kg of permanent magnets for EVs amount to a global warming potential (GWP) of 31.60 kg CO_2e, and ecological scarcity indicator of $9.98 \cdot e^{+4}$ UBP [73],[74]. These values can be attributed to the REE materials that, despite its low percentage in composition, can significantly increase the environmental impacts of Nd-Fe-B magnets. For instance, each kg of dysprosium presents a total eco-cost of € 1306.58, and a GWP impact of 121.15 kg of CO_2e, and neodymium € 222.60 and 35.73 kg CO_2e respectively, while in comparison the values for another key material such as boron amount to € 10.93 and 48.64 kg CO_2e. The eco-costs and global warming potential for electrolytic iron have not been found in the database [80]. Despite the low percentage of 3.96% in weight composition of the virgin Nd-Fe-B magnet, the impacts and associated costs for dysprosium are significant, further justifying the pursuit of recovering this critical material from EoL magnets.

In the study by Yang *et al*. [29], they assess that most EoL magnets go to shredders in a mixed way, increasing the complexity associated with magnet-to-magnet recycling as compositions will become contaminated. They have additionally discussed several problems that need to be addressed in the recovery and recycling of magnets, such as the presence of adhesive glues, metallic and epoxy coatings, plus the magnetic field presence.

As there are waste magnets also present varied sizes and weights besides composition, no universal recycling process can be designed [26]. Furthermore, as there is no standardization or labelling of PM materials that allow for easy classification of magnets, which often leads to contamination that can degrade the magnetic properties of the recovered material, hindering subsequent use [29], [30].

The challenges of coating removal prior to permanent magnet recycling have been studied by Couto *et al*. [26], among which, it was pointed out that zinc metallic coatings found in e-scooter PMs cannot be separated from the decrepitated powder by sieving, as it also forms a fine powder. To address that, they have successfully investigated the coating dissolution with ammonia-based solutions. It can be concluded then, that one of the major challenges to achieve the hydrogen decrepitation of magnets embedded in rotors is the presence of a metallic alloy.

In the case of zinc-based alloys, the restriction can hinder the process not only in regards to its start, but also by contaminating the resulting powder. In contrast, as previously mentioned, the same is not possible for removing nickel coatings due to kinetic restrictions. The alternative way of dissolving nickel-based coatings is, however, dependent on toxic solvents. It is in fact possible to remove the large flakes of nickel by sieving after hydrogen decrepitation is completed, but a key condition is that the nickel coating is not dense enough to hinder or prevent the decrepitation altogether [26], [30]. Considering these aspects, the presence of a zinc-type of coating might present a greater challenge than a nickel-based coating, which could potentially be removed mechanically to some extent post decrepitation.

Such differences in coating removal requirements, combined with the fact information on EoL magnet structure and composition is frequently limited, configures challenges in implementing a standard short-loop recycling process to EV magnets. It is therefore critical to have technical data on the magnets being subjected to the decrepitation treatment. If the hydrogen decrepitation main objective is to enable the disassembly of magnets embedded in the rotor, the presence of a coating alloy might hinder the process to the point of rendering it unfeasible. On the other hand, if the intention is to subject the EV pre-disassembled magnets to hydrogen decrepitation to achieve magnet-to-magnet recycling, a new pre-treatment step for coating removal in industrial-scale will be required. This highlights the need for further studies on the most used types of magnets' alloy coatings for EV applications, including an evaluation of supplier technical data in large-scale, to enable researchers to target viable, scalable processes.

The disassembly of permanent magnets has been evaluated for a variety of EoL magnet scrap, including for EV synchronous rotors, and it has been successfully shown that hydrogen is effective in selective recovery of magnetic material [18]. Hydrogen decrepitation has the potential to improve the disassembly and recovery of PM from its containing structure, but this process is sensitive to initial material composition homogeneity [24], [25], [26], [27], [30], [62], [64], [65]. In that sense, if it is possible to

implement hydrogen decrepitation for magnet streams of known average compositions such as EVs, the use of HD can be highly advantageous.

2.11 Current Research Gaps

In EV applications, permanent magnets are commonly embedded in rotor pockets and strongly bonded to the laminated electrical steel structure using adhesive materials. This design is necessary to ensure safe operation under high rotational speeds, elevated temperatures, vibration, and mechanical loading. However, the same design features that ensure operational safety also increase the complexity of end-of-life disassembly and recycling. Removing PMs from the laminated rotor stack without damaging the magnet body, protective coating layers, or surrounding electrical steel remains a significant challenge. Damage to the coating may expose the NdFeB magnet to air and moisture, increasing the risk of oxidation and degradation of magnetic properties [24], [25], [26], [27], [29], [64], [65].

Current studies on EV rotor disassembly have mainly focused on manual, mechanical, thermal, or robotic approaches for removing magnets from rotor structures. These methods have demonstrated that adhesive bonding is one of the major obstacles to efficient and non-destructive magnet recovery. However, limited research has addressed whether hydrogen decrepitation can be applied directly to magnets while they remain embedded inside EV rotor pockets, without prior complete mechanical removal. Investigating this possibility is important because it could reduce the need for extensive mechanical disassembly, lower process complexity, and potentially decrease the release of degradation products associated with thermal adhesive removal.

Another important research gap concerns the effect of protective coatings and adhesive layers on hydrogen-assisted recovery of EV rotor magnets. For hydrogen decrepitation to occur, hydrogen must reach the Nd-rich grain-boundary phase and the $Nd_2Fe_{14}B$ matrix. However, EV magnets may be surrounded by metallic coatings, polymeric or epoxy coatings, and adhesive bonding layers, all of which can restrict hydrogen access to the magnet surface. Standard HD parameters developed for loose, damaged, or pre-disassembled magnets may therefore not be directly applicable to embedded EV rotor magnets. Further evaluation is needed to determine whether coating removal, surface activation, or other pre-treatment steps are required before hydrogen treatment [26], [40], [45], [75], [76].

A further gap is related to the circularity potential of the laminated electrical steel rotor stack. Most studies on permanent magnet recycling focus primarily on the recovery of rare-earth-containing magnet material, while the preservation, recovery, or reuse potential of the surrounding electrical steel structure is rarely considered together with magnet recovery. If hydrogen decrepitation of magnets inside the laminated rotor structure can be achieved under suitable temperature and pressure conditions, it may be possible to recover magnetic material while also evaluating the circularity potential of the electrical steel.

Additionally, magnets embedded in EV rotors have limited exposed surface area because they are surrounded by multiple barriers, including metallic coatings, epoxy coatings, and adhesive materials. Therefore, standard hydrogen-decrepitation pressure and temperature conditions may not be sufficient to initiate or complete decrepitation. To develop industrially relevant recycling routes for EV permanent magnets, processes should minimize disassembly steps, reduce oxidation risks, limit magnet damage, and preserve as much material value as possible.

Circularity of permanent magnets is critical in an era of clean-energy development and increasing concern over supply security, as highlighted by the EU Critical Raw Materials Act. Therefore, investigating hydrogen decrepitation for the recovery of embedded EV magnets has potential to support recycling targets and automotive industry needs through a resource-efficient and industrially relevant route. This thesis investigates hydrogen decrepitation of NdFeB permanent magnets while still embedded in rotor laminated-stack pockets, combining literature analysis, experimental evaluation, and decision-flowchart modelling.

The main objective of this thesis is to develop an experimentally informed decision-support model for assessing whether hydrogen decrepitation can be used to recover NdFeB permanent magnets from EV rotor pockets, while considering coating barriers, adhesive bonding, process conditions, oxidation risks, and the circularity potential of both the magnets and the laminated electrical steel structure.

3 Methodology

3.1 Methodological approach

This thesis employs a mixed-methods research approach combining targeted literature review, experimental investigation, material-flow mapping, and decision-support modelling. This approach was selected because the research problem involves both technical feasibility and broader circularity assessment. The methodology therefore integrates information from scientific literature, environmental databases, experimental observations, and decision-support tools to evaluate the potential of hydrogen decrepitation (HD) as a disassembly and recycling route for permanent magnets embedded in EV rotors.

Literature review was conducted to map existing technologies and allow for the evaluation and definition of experimental parameters range, in order to investigate circularity potential of EV permanent magnets. Based on the review literature it was concluded that hydrogen assisted techniques are the most environmentally friendly techniques for recycling magnetic materials. Hydrogen decrepitation specially, as a technique that should be applicable in mild temperature and pressure conditions, will be further evaluated in the present work. Experimental treatment and analysis were conducted to evaluate the recycling potential of the PMs embedded in rotors, and parameters have been selected for study based on literature findings and validation needs. A flowchart for the manufacturing of NdFeB magnets, from the sourcing of the main REE used to the final magnet product, was compiled through the combination of literature data and datasets available in Ecoinvent for mining and beneficiation, and permanent magnet production for EVs. An additional decision matrix was modelled to map the investigation flow during the experiments and to support informed-decisions for future business-models regarding PM circularity.

The literature review was conducted using scientific databases and search tools including Scopus/Elsevier, Google Scholar, SpringerLink, ResearchGate, and Connected Papers. Search terms included combinations such as “NdFeB AND recycling,” “permanent magnet AND recycling,” “permanent magnet AND circularity,” “NdFeB AND manufacturing,” “rare-earth element AND recycling,” “hydrogen-assisted recycling,” “hydrogen decrepitation AND permanent magnet,” “HDDR,” “permanent magnet AND disassembly,” and “electric vehicle AND permanent magnet AND disassembly.” Papers were selected based on relevance of title, keywords, abstract, and content, with priority given to peer-reviewed journal articles, review papers, book chapters, and conference papers. No strict publication-year limit was applied because fundamental work on permanent magnet manufacturing and hydrogen processing dates back several decades. However, when possible, priority was given to studies published from 2015 onwards to reflect recent developments in EV magnet recycling and circularity.

Literature review was used to investigate the available disassembly methods for EV magnets and future work focus, the current technologies available for PM recycling including their advantages and drawbacks, the specific methods, mechanism, and key

parameters for the chosen recycling methods, in other words hydrogen assisted techniques, and to allow the compilation of the manufacturing flowchart for EV Nd-Fe-B permanent magnets. The study follows a primarily deductive and iterative methodological logic. Existing theory and literature on hydrogen decrepitation were first used to define expected reaction behaviour and suitable process parameters. These assumptions were then tested experimentally using embedded EV rotor magnet samples. The results were compared with literature expectations and used to refine the decision-support framework. Because the number of experimental runs was limited by access to external laboratory facilities and equipment constraints, the experimental work should be understood as a feasibility investigation rather than a full process optimization study.

A comparative parameter analysis was conducted based on literature review and experimental analysis, by taking into consideration reaction enablers, inhibitors, potentially optimal values, and experimental findings. The main experimental objective is the evaluation of the viability of hydrogen decrepitation at mild conditions to recover magnetic powder from permanent magnets still embedded in EV rotor stack pockets. This study used a new rotor stack, with embedded Nd-Fe-B permanent magnets bound by adhesive, with an expected polymeric protective coating. Specific initial conditions of the magnet, such as detailed composition and all coating layer types were not know in full prior to the experimental analysis. The magnet's source has not been disclosed and will not be considered for the purposes of this research.

The hydrogen decrepitation experiments were conducted in and external partner laboratory, and due to limited access to laboratory facilities, the number of experimental runs was restricted. This limitation led the experimental approach to be adjusted on an-ongoing basis targeting full decrepitation of the material and powder recovery by testing from the mildest conditions to the most extreme possibilities within the defined parameter range. The hydrogen decrepitation has been qualitatively verified by X-ray diffraction (XRD) analysis. Differential Scanning Calorimetry (DSC) analysis was also used to evaluate the glass transition and degradation temperatures of the adhesive material binding the laminated rotor stack, in order to assess whether hydrogen treatment could be performed while limiting adhesive degradation and potential release of harmful fumes.

According to literature, full magnet decrepitation, regardless of fine powder fraction yield, should be possible in mild conditions, for instance room temperature and hydrogen pressure of 0.5 to 1.0 bar. However, literature studies start from pre-disassembled magnets, that present from little damage to fully fractured body, either caused from the disassembly process or on purpose to allow for greater surface reaction area. This shall not be the case for the magnets in this study, which will be in a new, integral state, with no prior surface damage, and with limited exposed surface due to its embedded characteristic. In this sense, hydrogen decrepitation at mild conditions will be evaluated in face of such surface limitations.

3.2 Experimental set-up and analysis

The samples were obtained from a new EV rotor provided by TRATON Group R&D team. The rotor was supplied in integral condition and consisted of a laminated stack containing embedded permanent magnets. The magnets were bounded using a strong adhesive of unknown composition and were protected by an external coating. No prior information regarding the presence or composition of additional alloy coatings was available before the experiments.

The adhesive's glass transition point has been evaluated by DSC analysis performed using a Netzsch DSC 214 Polyma equipment. Duplicate samples were weighted in a Metler Toledo precision scale and weighted 18.6 mg and 21.2 mg. Measurements were conducted in nitrogen atmosphere with an initial temperature (T_0) of 20°C, heating rate of 10°C/min, and acquisition rate of 300 points/min. The samples were heated until reaching a temperature (T_i) of 290°C, where the system was kept in isothermal state for 15 minutes. After the isothermal step was concluded, temperature was increased up to (T_f) 300°C. Analysis was conducted in the Netzsch SmartMode 8.0 software.

Due to size limitations of the hydrogen treatment equipment, the laminated stack was sectioned into slices with equal dimensions, symmetry, and number of magnets, henceforth named Sample 1 (S1), Sample 2 (S2), Sample 3 (S3), Sample 4 (S4). A safety assessment was conducted by the responsible technical partner team before cutting, ensuring that no magnets were directly cut during sample preparation.

The HD experiments were performed at an external partner laboratory. The initial experimental setup consisted of a tube furnace with a uniform heating zone up to 1000°C and hydrogen gas flow input of 5 l/min. The system's operating pressure was limited to 1 bar. After treatment, the samples were cooled to room temperature under an inert argon atmosphere. The selected experimental parameters and corresponding sample preparation conditions used in this study are summarized in **Table 5**.

Magnetic flux measurements were performed before and after hydrogenation using an EMF detector. They were conducted at a fixed distance of 40 cm to allow comparison.

Table 5: Experimental parameter selection and surface-preparation conditions used to evaluate hydrogen decrepitation of embedded Nd-Fe-B magnets.

Sample	Temperature (°C)	Pressure (bar)	Exposure time (h)	Extra preparation
S1	170 ± 5	1	4	None
S2	170 ± 5	1	4	Surface scratching
S3	300 ± 5	1	6	Complete top surface removal

Phase characterization was performed using powder X-ray diffraction (XRD), with a Rigaku MiniFlex X-ray radiation diffractometer to evaluate hydrogenation and decrepitation. Measurements were conducted in θ/2θ scan mode using a step width of 0.005°, scan speed of 0.10°/min, and a scan range of 20-90°. This angular range was selected because Nd-Fe-B and hydride phases reported in the literature are expected within this interval. A voltage of 40kV and current of 15mA were used for the scans.

Prior to analysis, the recovered material was manually ground using a mortar and pestle, and a thin powder layer was immediately placed on the sample holder. The grounding step was not conducted in inert atmosphere as the sample could not be recovered from the partner laboratory stored with inert atmosphere control. Phase identification was then used to evaluate the presence of remaining $Nd_2Fe_{14}B$ phases and hydrogenated compounds formed during treatment.

4 Results

4.1.1 Experimental flow diagrams

The experimental decision matrixes developed in this work, and presented in **Figures 12**, **13, and 14**, were created to support the evaluation of technical feasibility of HD as a disassembly and recycling method for embedded EV rotor magnets. The diagrams summarize the main experimental decision paths, parameter considerations, and technical constraints identified during the study.

Figure 12 presents the sample preparation decision matrix, including considerations related to rotor sectioning, demagnetization, and magnet accessibility. **Figure 13** summarizes the hydrogen decrepitation pre-treatment evaluation, focusing on coating presence, surface preparation, and their influence on hydrogen diffusion and decrepitation effectiveness. Finally, **Figure 14** presents the overall experimental feasibility investigation matrix, integrating the main process evaluation steps, material recovery considerations, and potential recycling routes assessed throughout the study. Together, they provide a structured framework for assessing the feasibility of hydrogen-assisted recycling routes for EV rotor magnets.

4.2 Key environmental impacts of Nd-Fe-B magnets

Environmental eco-cost data for the main Nd-Fe-B magnet components were compiled from the Idemat 2026 database Revision A1 and are presented in **Figure 15**. The values correspond to upstream and downstream environmental costs associated with the production of 1kg of each material in metallic form. **Figure 16** presents the eco-cost comparison for different grades of Nd-Fe-B magnets, including virgin and recycled material routes.

To support the evaluation of the environmental impacts associated with Nd-Fe-B magnet production, a process flowchart was developed based on available literature and databases, as presented in **Figure 17**. The flowchart summarizes the main unit operations involved in the extraction, processing, and manufacturing of RE-based magnets. It presents the overall process route from rare-earth extraction to final magnet production. The process includes multiple hydrometallurgical and pyrometallurgical operations associated with REE beneficiation and magnet production. These operations involve the consumption of water, energy, and chemical reagents, while also generating process waste streams.

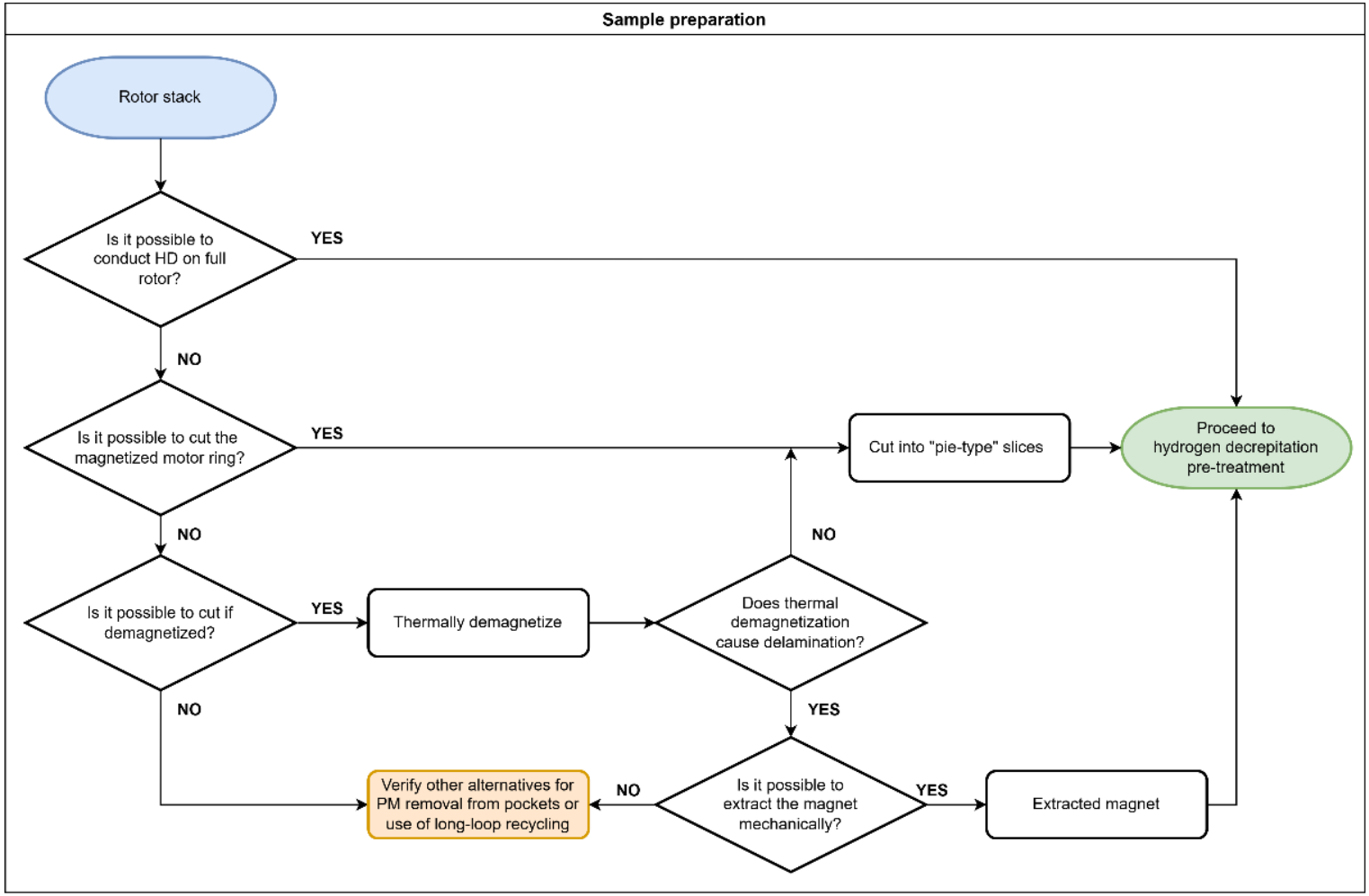


Figure 12: Sample preparation decision flowchart for embedded Nd-Fe-B magnets in EV rotor stack sections.

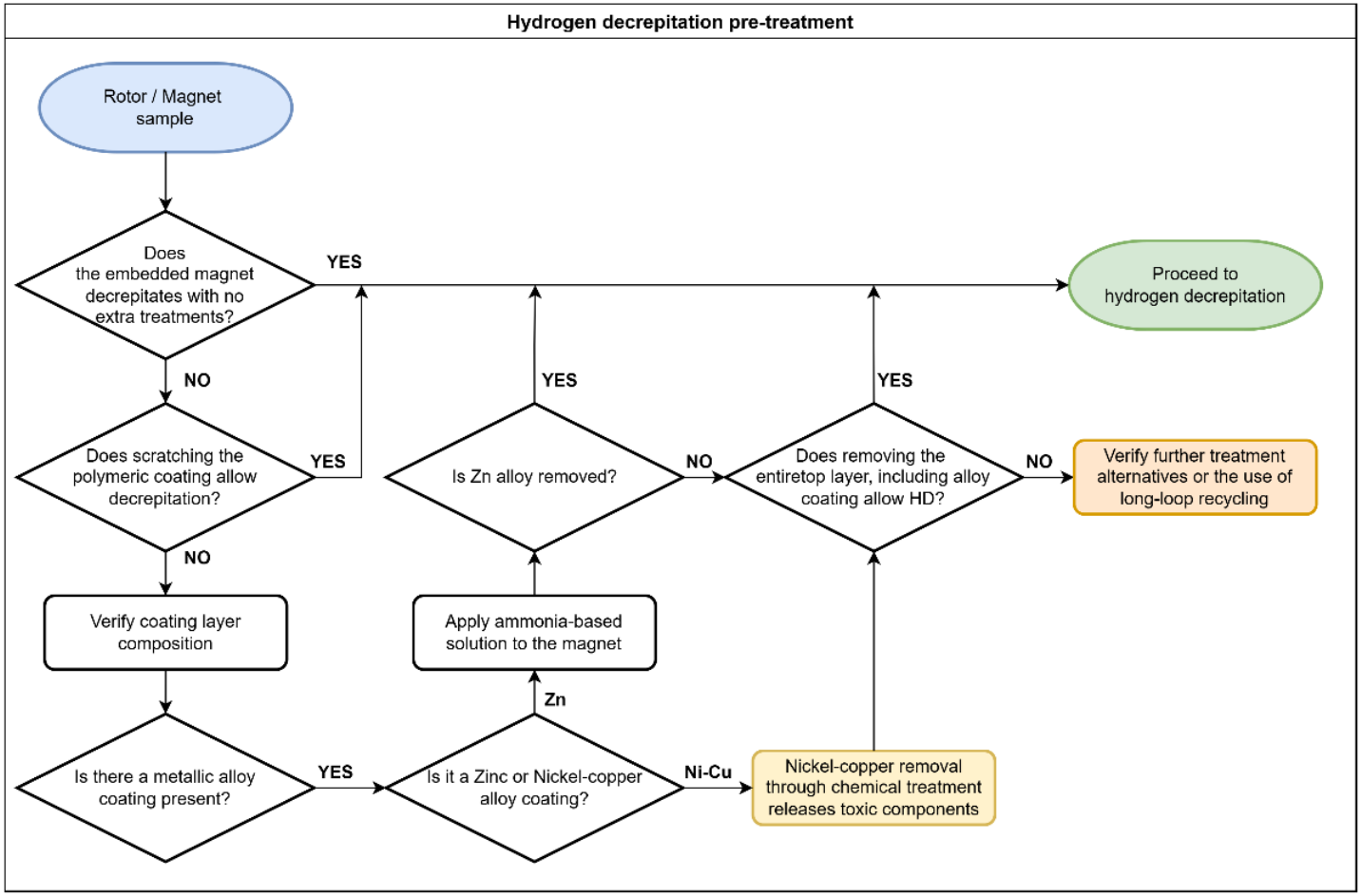


Figure 13: Hydrogen decrepitation pre-treatment decision flowchart based on coating condition and surface accessibility.

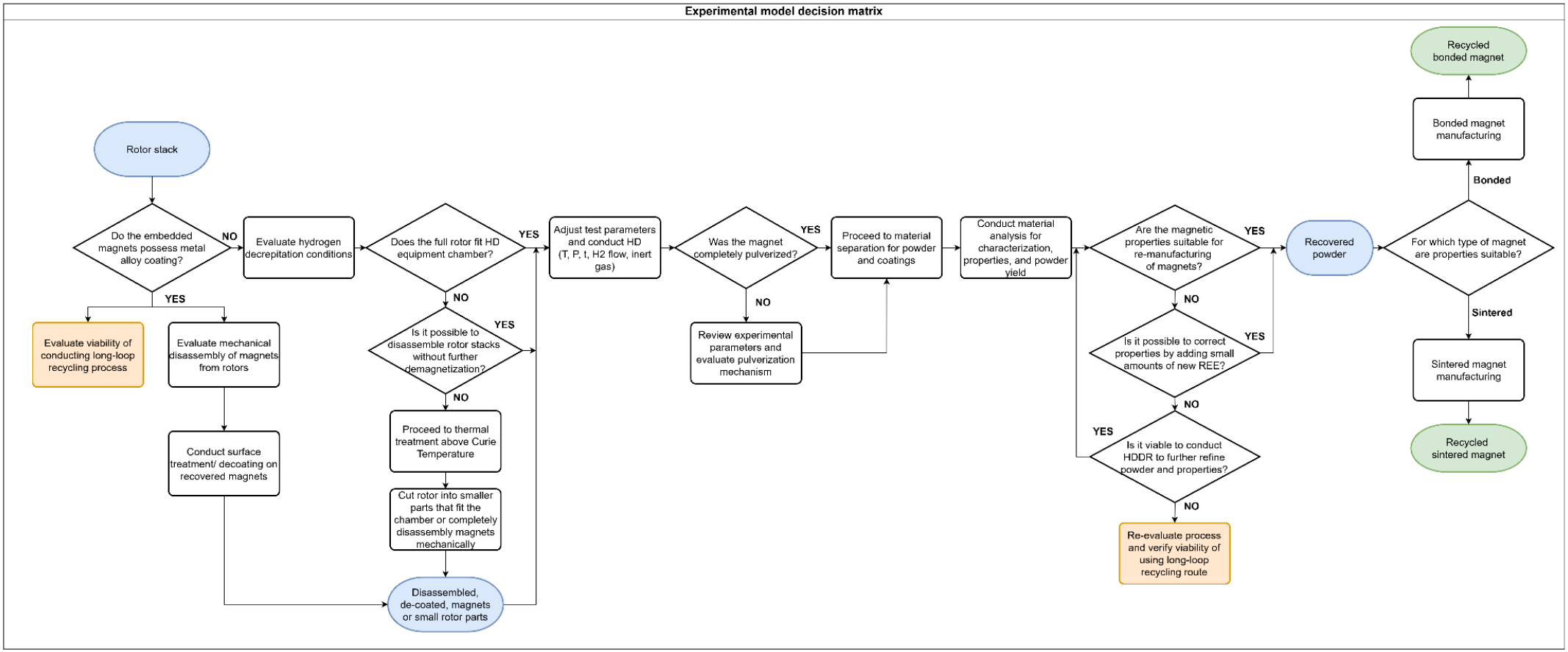


Figure 14: Experimental feasibility investigation decision flowchart for hydrogen decrepitation outcomes and recycling-route suitability.

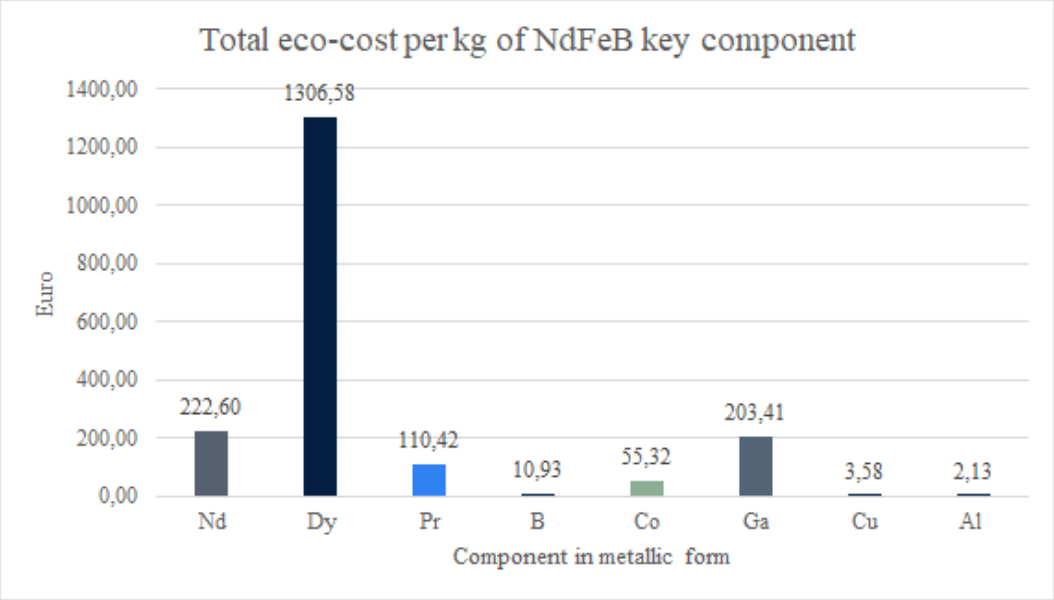


Figure 15: Eco-costs of key Nd-Fe-B permanent magnet components in metallic form, based on upstream and downstream background processes. Compiled from Idemat2026 [80].

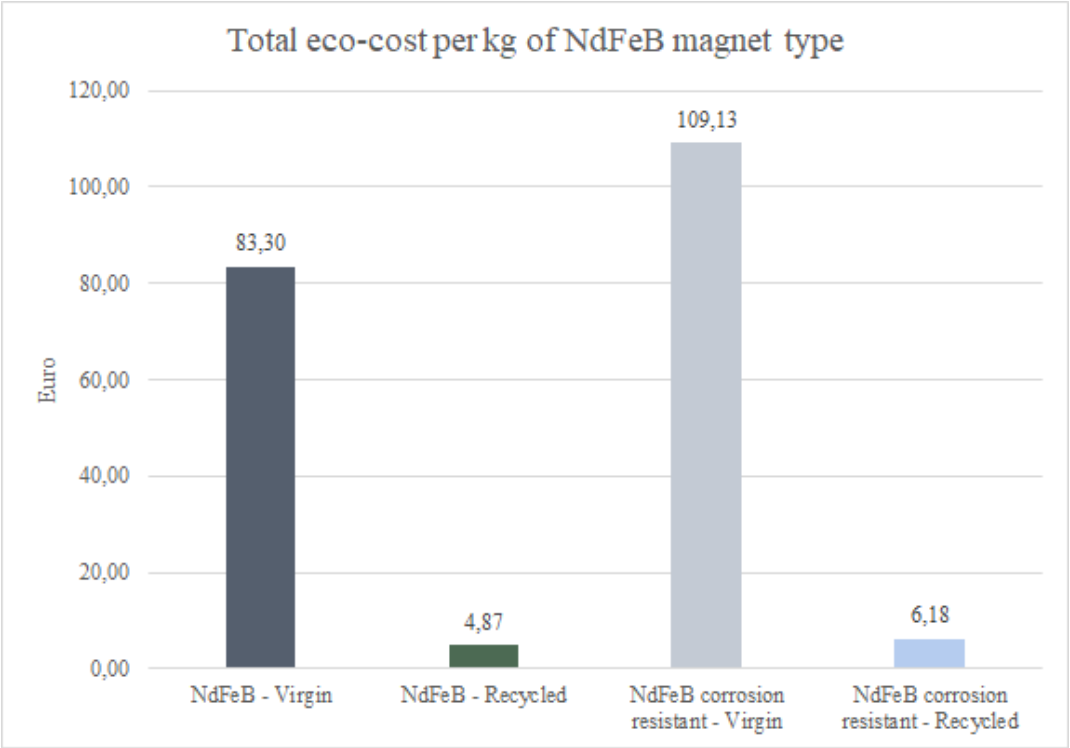


Figure 16: Eco-costs of different Nd-Fe-B permanent magnet grades. Compiled from Idemat2026 [80].

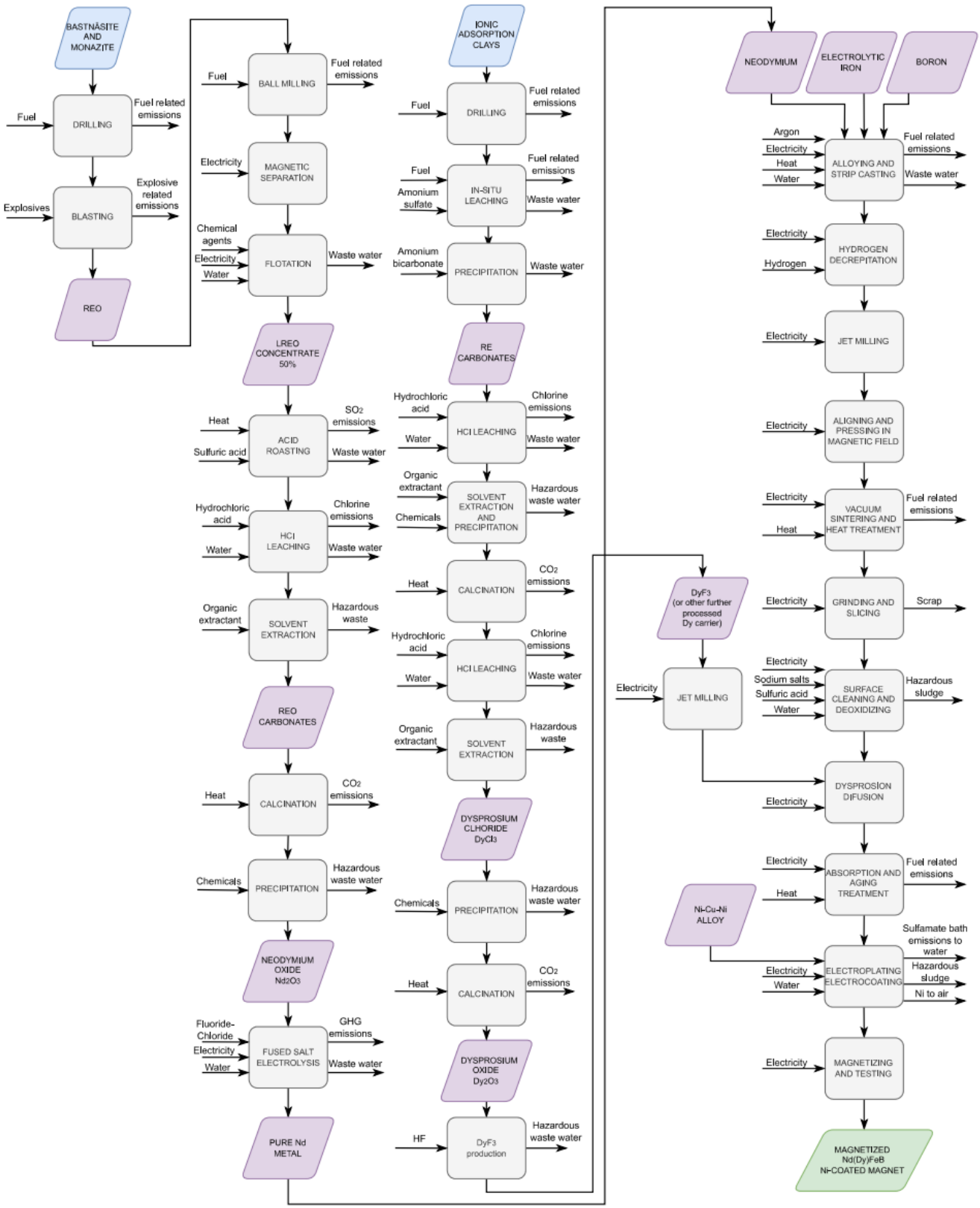


Figure 17: NdFeB permanent magnets manufacturing process flow, from key REE extraction to final magnet. Developed based on literature and database sources [40], [41], [42], [49], [50], [53], [81].

4.3 Evaluation of HD treatment of embedded magnets

Prior to HD treatment, DSC analysis was conducted to evaluate the thermal behaviour of the adhesive material binding the laminated rotor stack containing the magnets. The analysis identified a glass transition temperature of 132.6°C and 132.7°C, and the onset of material degradation at approximately 220°C for both samples.

The cut segments of the laminated stack (S1, S2, S3, S4) underwent HD treatment under different experimental parameters and the hydrogenation results obtained for each sample are summarized in **Table 6**.

Table 6: Summary of hydrogen decrepitation results for each sample.

Sample	Results and observations
S1	No decrepitation, pulverization, volume expansion, or magnetic flux loss was observed.
S2	No decrepitation, pulverization, volume expansion, or magnetic flux loss was observed.
S3	Reaction was observed on the side where coatings were removed. Further confirmation of hydrogenation necessary through XRD analysis.

No visible decrepitation, pulverization, volume expansion, or reduction in magnetic flux was observed after treatment. A slight colour change in the adhesive material binding the magnets to the rotor laminated stack was observed after thermal exposure. Cross-sectional analysis of the magnet identified the presence of a metallic alloy coating.

S2 was treated under the same temperature, pressure, and exposure conditions as S1. Prior to treatment, deep scratches were introduced into the coating surface of the magnet on one side of the sample. No visible decrepitation, pulverization, volume expansion, or significant reduction in magnetic flux were observed.

To evaluate hydrogen decrepitation of embedded magnets with limited exposed surfaces, the coating on one side of S3 was completely removed through drilling. The sample was immediately subjected to hydrogenation treatment to minimize air exposure. Treatment was conducted at 300°C for 6 hours under 1 bar hydrogen pressure.

Visible reaction occurred on the side where the coating had been removed. Degradation of the adhesive material binding the laminated stack was also observed after treatment. Volume expansion occurred on the magnet with exposed surface, leading to partial disassembly of the laminated stack. Fracture formation was observed in the reacted region of the magnet. However, complete pulverization into fine powder was not achieved.

Magnetic flux measurements performed before and after treatment indicated a reduction of approximately 85% in magnetic field intensity.

To evaluate the effectiveness of hydrogenation, powder XRD analysis was conducted on powder recovered from different depths of the magnet. The phase identification results are presented in **Figure 18**. The XRD analysis identified the presence of both $Nd_2Fe_{14}B$ and the hydrogenated $Nd_2Fe_{14}BH_{1.04}$ phases in the recovered material. The identified phases correspond to the International Centre for Diffraction Data (ICDD) database codes (DB-cards) 01-090-3538 and 01-086-0274 respectively.

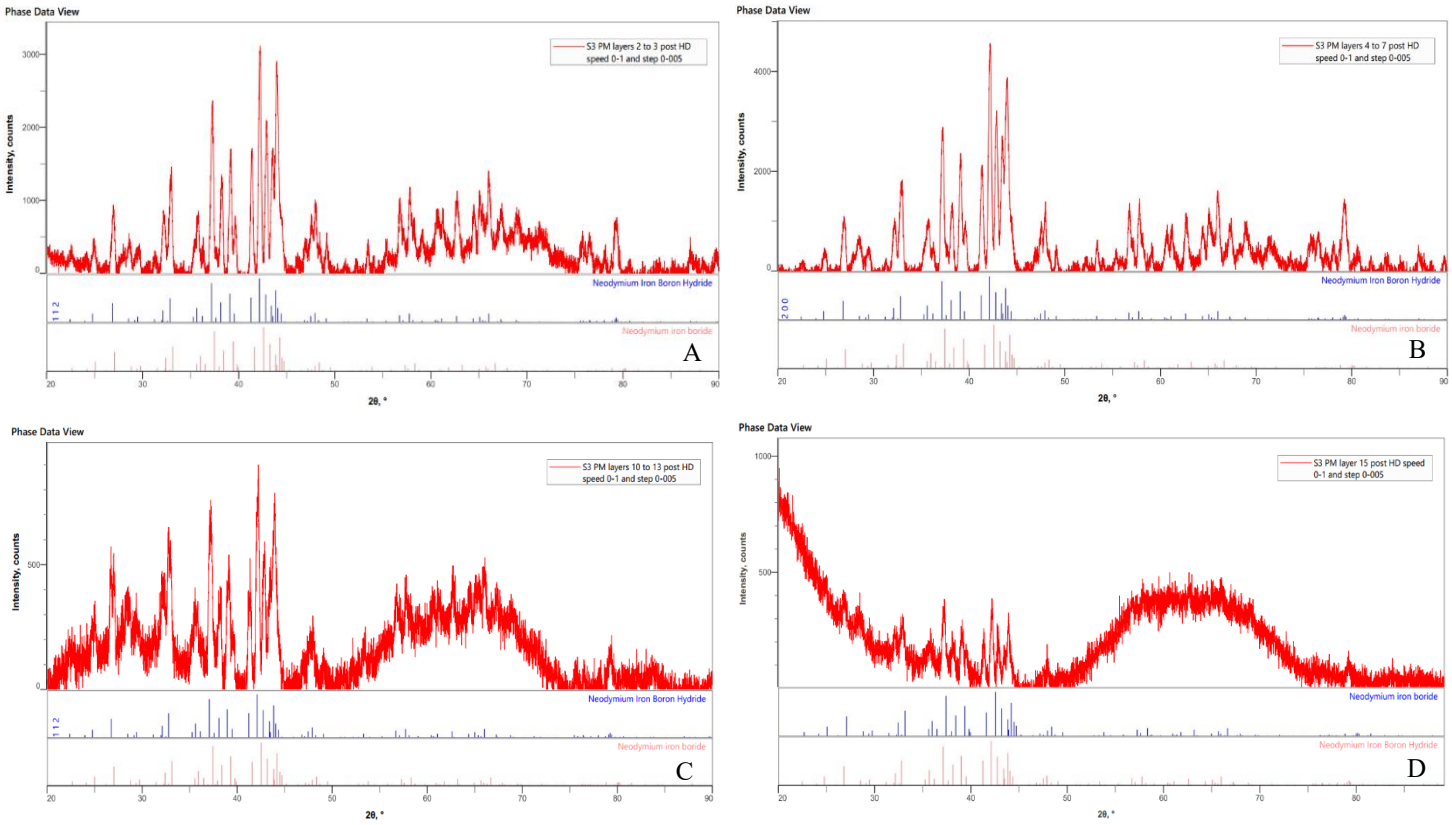


Figure 18: Powder XRD analysis and phase identification at different layers from Sample 3 post hydrogenation treatment, where layers are: (a) 2-3 (top), (b) 4-7, (c) 10-13, (d) 15 (middle length)

4.4 Business model proposal

As an output of this thesis, the decision matrices presented in **Figure 19** and **Figure 20** were developed to support technical and strategic evaluation of EoL EV Nd-Fe-B magnet and critical materials recycling pathways.

The proposed framework integrates considerations related to disassembly, recycling route selection, hydrogen-assisted processing, and material recovery strategies. The matrices

additionally distinguish between short-loop and long-loop recycling approaches and include process considerations associated with scalability, contamination risks, and material retention.

Table 7 complements the proposed framework by summarizing selected recycling pathways according to circularity potential, technology maturity, process complexity, environmental perspective, and route objective. The classifications were developed through qualitative literature comparison and will be further discussed in Chapter 5.3.1.2.

Table 7: Qualitative strategic assessment of selected Nd-Fe-B permanent magnet recycling pathways.

Recycling route	Circularity potential	Technology maturity	Process complexity	Environm. perspective	Route Objective	Ref
Hydrogen decrepitation (HD)	High	High	Low	Very favourable	Magnet powder recovery and alloy preservation	[62], [63], [69]
HDDR	Very high	High	Low	Very favourable	Magnet powder recovery with enhanced material properties	[62], [63], [69]
Complete leaching	Moderate	High	Low	Moderate	Mixed REO product recovery	[62], [63], [69]
Selective leaching	Moderate	High	Low	Moderate	Selective REE recovery	[62], [63], [69]
Molten-salt electrolysis	Moderate	High	Low	Favourable	High-purity REE recovery	[62], [63], [69]
Ionic-liquid extraction	High	Low	Low	Promissing	Advanced REE separation	[62], [63], [68], [69]

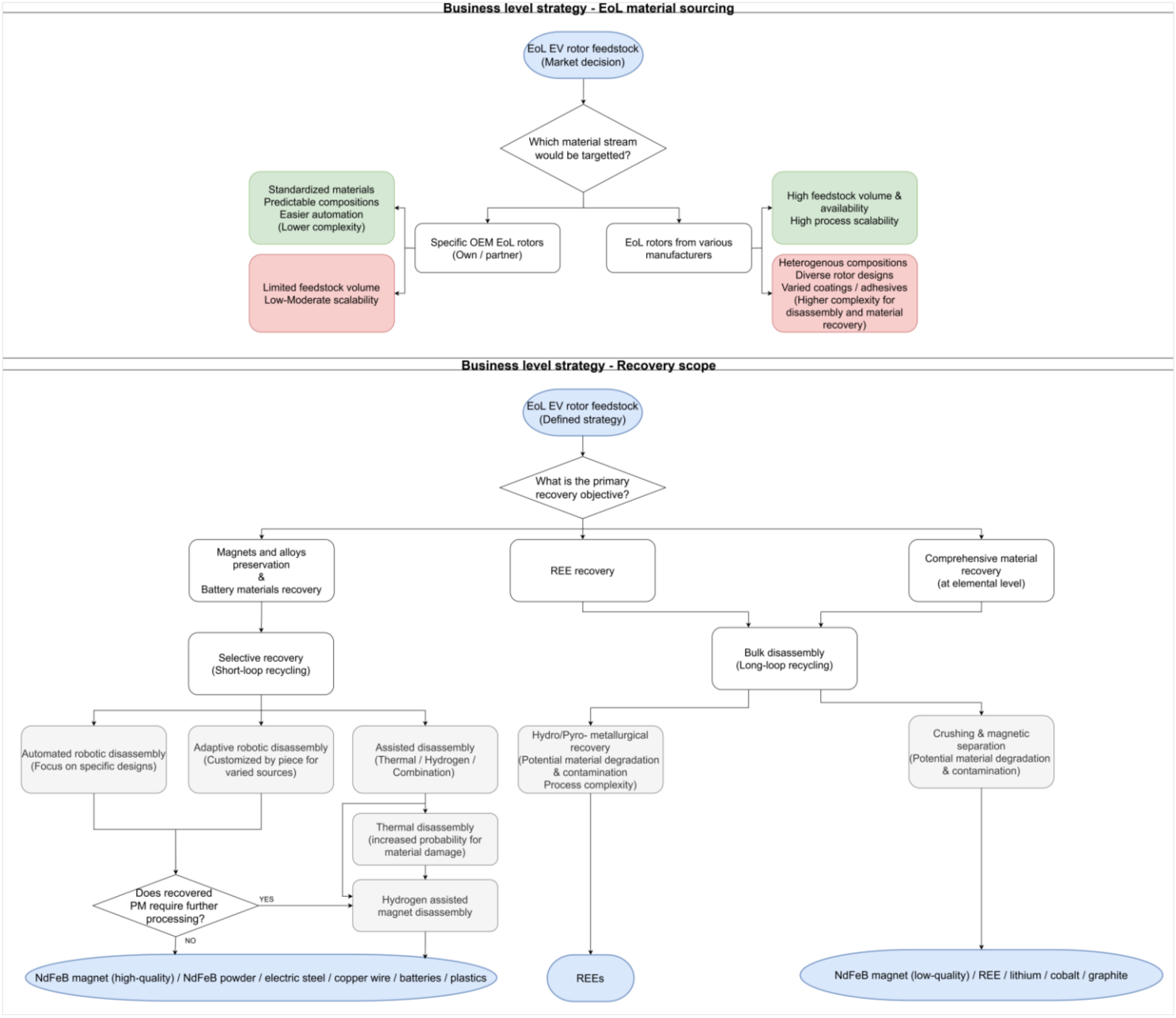


Figure 19: Strategy decision framework at business level for EoL EV and permanent magnet circularity.

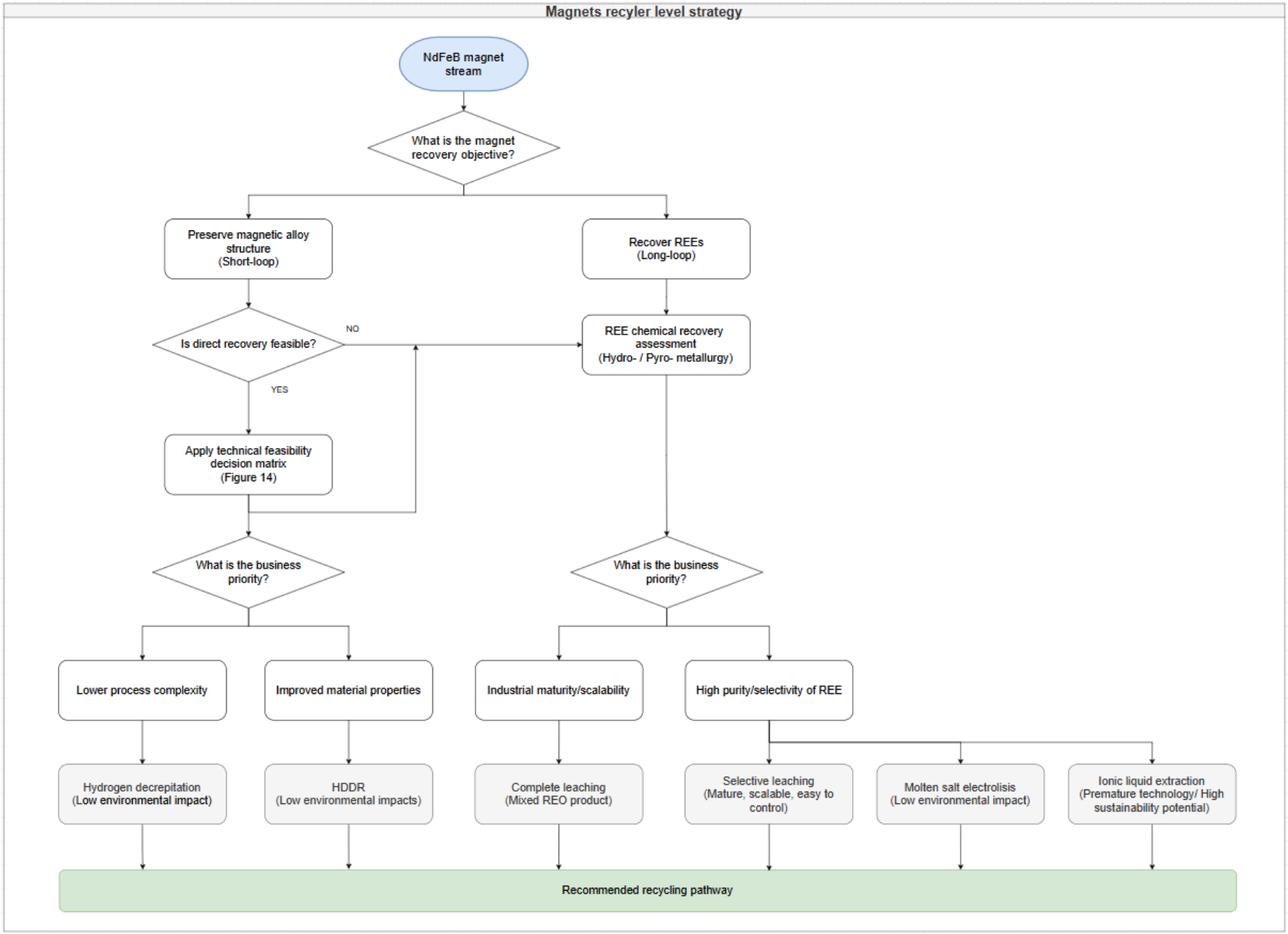


Figure 20: Strategy decision framework at recycler level for EoL EV and permanent magnet circularity.

5 Discussion

5.1 Environmental relevance and database limitations in Nd-Fe-B magnet production

The eco-cost assessment compiled from the Idemat 2026 database showed that REE Dy and Nd contribute substantially more environmental impact per kilogram than other magnet constituents, as shown in **Figure 15**, except for Ga. However, composition-wise, Ga represents approximately 0.2% of the magnet, while Dy and Nd amount to 4.0 - 6.1% and 18 - 21.6% respectively, leading to lower overall contribution to the magnet's eco-costs. Additionally, the comparison between virgin and recycled magnets demonstrated a reduction in eco-costs of more than 94% for recycled materials across the evaluated magnet grades, as presented in **Figure 16**. These results highlight the environmental relevance of RE recovery and recycling for reducing resource depletion and supporting supply security.

The environmental assessment conducted in this study highlighted the complexity of Nd-Fe-B magnet manufacturing and the challenges associated with evaluating its environmental impacts. The process flow analysis presented in **Figure 17** showed that the production of RE-based PMs depends on multiple hydrometallurgical and pyrometallurgical operations with high consumption of energy, water, and chemical reagents. Significant environmental hotspots were identified in leaching, solvent extraction, and precipitation stages due to their intensive reagent use and waste generation. These observations reinforce the importance of developing circularity strategies for Nd-Fe-B magnets.

During the assessment of Nd-Fe-B magnet manufacturing routes, several inconsistencies and limitations were identified in both literature and environmental databases. Backtracking of Dy_2O_3 production route in the EcoInvent database linked dysprosium production to HREE extraction from mineral ores in China. However, industrial HREE production in China is primarily associated with IAC deposits. This discrepancy demonstrates the difficulty in accurately tracing industrial production routes through available databases.

Different literature sources also described multiple approaches for Dy diffusion into the Nd-Fe-B magnets, including electrochemical diffusion using Dy_2O_3, spraying of DyF_3, and metallic dysprosium deposition. Although all approaches aim to improve thermal stability and magnetic performance, the reported processing routes and efficiencies vary significantly. Information regarding the specific industrially adopted process was limited in both literature and environmental datasets. Based on supplier information, grain-boundary diffusion using DyF_3 appears to be the most commonly applied industrial route.

Additional limitations were identified in the Idemat 2026 database. Environmental impact data were only available for the metallic RE forms, while information for compounds such

as DyF_3 and Dy_2O_3 was not available. This restricts the environmental comparison of different doping components and processing routes used in high-performance magnet manufacturing.

Another limitation identified during the environmental assessment was the lack of standardized manufacturing process information. Environmental datasets for recycled magnets are commonly based on combinations of literature data, industrial datasets, and manufacturer-specific primary information. Since manufacturing routes, process efficiencies, reagent consumption, and waste generation can vary significantly between producers, the environmental impacts reported in literature and databases may not fully represent industrial reality. This uncertainty complicates targeted process optimization strategies.

5.2 Hydrogen decrepitation viability as a disassembly method for embedded EV rotor magnets

5.2.1 Assumptions and parameters definition

The experimental parameters were selected to evaluate the feasibility of HD for the disassembly and recovery of embedded Nd-Fe-B magnets under relatively mild operating conditions. The study focused primarily on achieving magnet hydrogenation and pulverization for powder recovery. Dehydrogenation and detailed powder characterization were considered outside the scope of this work due to equipment limitations and are proposed for future studies.

The selected parameters considered the combined influence of temperature, pressure, reaction time, and coating presence on hydrogen diffusion and pulverization effectiveness. Since these factors are interdependent, the parameter ranges were chosen to balance reaction effectiveness, processing time, and operational limitations of the available equipment.

Temperature selection

The main experimental temperature range was defined between approximately 170°C and 300°C. The lower limit was selected to remain below degradation temperature of the adhesive material used in the laminated stack, avoiding the release of potentially hazardous fumes during treatment. The upper limit was selected to avoid excessive grain growth and reduced pulverization effectiveness at higher temperatures. The selected temperature range also considered the need to promote hydrogen diffusion and magnet pulverization while limiting excessively long reaction times. Room temperature conditions were additionally evaluated to investigate the initial hydrogen penetration behaviour before the formation of the hydrogenated surface layers that could hinder further diffusion into the magnet bulk.

Pressure effects:

The pressure range was selected between 1 and 5 bar H_2, based on equipment limitations and the objective of improving hydrogen penetration and reaction initiation. Despite the expected increase in reaction time and reduced hydrogen retention at elevated temperatures at lower pressures, 1 bar was selected due to the initial equipment used.

Higher pressures were evaluated to promote faster surface activation and improve hydrogen diffusion into the embedded magnets. However, the study remained focused on relatively mild operating conditions to assess the practical feasibility of HD as a potential disassembly and recycling approach for EV rotor magnets.

Reaction time selection:

Initial exposure times were defined between 4 to 6 hours to evaluate the progression of hydrogenation and pulverization under different parameter combinations. Longer reaction times were considered necessary due to the presence of embedded magnets, adhesive materials, and potential coating barriers that could restrict hydrogen diffusion into the magnet bulk.

Since industrial implementation requires reduced processing times and stable material recovery, the experiments also aimed to evaluate whether effective decrepitation could be achieved under comparatively mild conditions and within practical treatment durations.

Coating effects and surface preparation

The magnets presented external protective coatings of unknown composition, which were initially not expected to influence hydrogen diffusion and decrepitation effectiveness. Since no prior information regarding the coating type or thickness was available, the samples were evaluated in different surface conditions to assess the influence of coating presence and exposed magnet surfaces on the treatment performance.

Additional surface preparation was performed in selected samples to evaluate whether increased surface exposure could improve hydrogen penetration and facilitate pulverization of the embedded magnets.

5.2.2 Experimental feasibility investigation

The decision matrices presented in **Figures 12**, **13**, and **14** were developed through an iterative process combining literature review and experimental observations. Initially, key technical factors influencing the HD process, magnet recovery, coating behaviour, and recycling route selection were identified from published studies. During the experimental phase, additional decision points were incorporated to account for practical constraints found during sample preparation and hydrogen treatment, including equipment limitations, unknown coating presence and compositions, and incomplete decrepitation. The resulting matrices therefore represent both established knowledge from the literature and insights generated during the experimental investigation.

Figure 12 presents the sample preparation decision matrix. The framework was developed to evaluate the practical feasibility of preparing EV rotor samples for hydrogen decrepitation treatment. The matrix consists of equipment limitations, rotor dimensions, magnetization state, and potential disassembly strategies. It provides a structured sequence of decisions that guide the preparation of rotor samples while minimizing unnecessary processing steps and preserving the integrity of embedded magnets.

Figure 13 presents the HD pre-treatment decision matrix. The framework was developed to evaluate potential surface-related barriers to hydrogen diffusion and decrepitation, particularly the presence of protective coatings. Based on the observed response of the magnet to hydrogen exposure, the matrix guides the selection of additional pre-treatment steps, including coating characterization, coating removal, or alternative recycling pathways. This framework aims to identify conditions under which HD can be successfully applied while minimizing unnecessary material processing.

Figure 14 represents a structured decision-support framework that links experimental outcomes with circularity objectives. By integrating process feasibility, material quality assessment, and remanufacturing potential, the framework provides a methodology for evaluating whether hydrogen-assisted disassembly and recycling can be implemented as a viable short-loop route for embedded EV rotor magnets.

5.2.3 HD treatment results

The experimental investigation demonstrated that hydrogen decrepitation of embedded Nd-Fe-B magnets is possible under specific conditions, although complete magnet disassembly from the rotor pockets was not achieved. The main limitations identified during the experiments were related to restricted surface availability caused by adhesive bonding and the presence of metallic protective coatings.

S1 and S2 did not present visible decrepitation, pulverization, volume expansion, or significant magnetic flux reduction after hydrogen treatment. The absence of reaction in both samples suggests that the metallic coating layer acted as a barrier to hydrogen diffusion into the magnet bulk.

S2 differed from S1 by the presence of deep scratches introduced into the coating surface before HD treatment. However, localized scratching was insufficient to initiate decrepitation, indicating that limited surface exposure alone may not provide adequate hydrogen access to the magnet structure under the evaluated conditions and time limitation.

In S3, decrepitation was observed after complete removal of the coating layer from one exposed magnet surface. The sample also presented visible volume expansion, cracking, and significant reduction in magnetic field intensity after treatment. The decrease in magnetic flux of approximately 85-86% indicates substantial demagnetization consistent with hydrogenation of the magnetic material.

The occurrence of hydrogenation in S3 was confirmed through XRD analysis, which identified both $Nd_2Fe_{14}B$ and $Nd_2Fe_{14}BH_{1.04}$ phases in the recovered material, as shown in **Figure 18**. However, complete pulverization into fine powder was not achieved, and additional characterization is still required to evaluate material yield, phase conversion extent, and magnetic properties of the recovered powder, in accordance with the proposed framework in **Figure 14**.

The experimental results also indicate that the hydrogen decrepitation behaviour of embedded magnets differs from previously reported results for already-disassembled magnets. In previous studies, decrepitation was reported at lower pressures and temperatures even in magnets containing metallic coatings. However, those samples often presented exposed surfaces or coating damage from prior disassembly operations. The present study suggests that limited surface accessibility plays a significant role in restricting hydrogen penetration into embedded rotor magnets. Further evaluation is required for EoL recovered rotors presenting degradation from operation conditions or dismantling. Furthermore, additional studies are needed to evaluate the effectiveness of disassembly of inner rotor laminated stacks, where surface availability is even more limited and targeted coating removal is not possible.

5.2.3.1 Effect of coating and surface accessibility

The experimental results suggest that the metallic coating layer significantly influenced hydrogen diffusion and decrepitation effectiveness. Although polymeric coatings alone have previously been shown to allow hydrogen decrepitation, the additional intact metallic alloy layer observed in the present samples likely acted as a diffusion barrier.

Previous studies investigating hydrogen behaviour in nickel-, copper-, and nickel-copper-based alloys have shown that these materials can present hydrogen trapping behaviour and increased resistance to hydrogen-driven embrittlement. Since embrittlement and fracture contribute to the decrepitation mechanism, the coating layer may have reduced hydrogen penetration and prevented crack propagation into the magnet bulk.

Complete removal of the coating layer in S3 enabled decrepitation to occur, while untreated and scratched samples did not react. This observation indicates that coating continuity and exposed surface area are critical parameters for hydrogen-assisted disassembly of embedded magnets.

However, the exact composition and thickness of the coating used in the rotor magnets were unknown. Therefore, the behaviour observed in this study cannot be generalized to all Nd-Fe-B magnet coating systems. Further investigation of hydrogen diffusion through commercially used magnet coatings is required to determine the pressure and temperature conditions necessary to enable hydrogen penetration and industrial scale decrepitation.

5.2.3.2 Partial decrepitation after coating removal

Although decrepitation occurred after complete removal in S3, the magnet did not fully pulverize into fine powder. The recovered material presented a friable structure with visible cracking and expansion, but large portions of the magnet remained mechanically connected.

One possible explanation is that the selected pressure and temperature conditions were insufficient to promote complete hydrogen penetration and stress development throughout the entire magnet volume. The experiments on S1, S2, and S3 were limited to 1 bar hydrogen pressure due to equipment constraints, restricting the evaluation of pressure effects on decrepitations efficiency.

Another possible explanation is related to the elevated treatment temperature of 300°C. Higher temperatures may reduce internal stress accumulation within the magnet by promoting hydrogen desorption, thereby reducing the effectiveness of the pulverization mechanism. Additionally, the formation of hydrogenated hard surface layers – NdH_x – could have partially hindered further hydrogen diffusion into the magnet bulk during treatment.

Finally, the limitations previously discussed might have led to increased reaction time. With the hydrogen treatment duration being limited to evaluate feasibility within specified timeframe, hydrogenation might not have been concluded.

Despite incomplete pulverization that hindered disassembly, XRD analysis confirmed that hydrogenation occurred throughout different magnet layers, indicating that hydrogen diffusion was not limited exclusively to the outer surface regions.

5.2.4 Implications for industrial feasibility

Several practical limitations identified during the experiments are relevant for future industrial-scale implementation of hydrogen-assisted recycling routes for EV rotor magnets.

The experimental equipment presented significant sample-size limitations, requiring the rotor stack to be sectioned into smaller segments prior to treatment. In addition, the lack of pressure control restricted the evaluation of optimized hydrogenation conditions and their influence on reaction time and pulverization effectiveness.

The condition of EoL materials also represents an important consideration for industrial applications. Real EoL rotor magnets may present corrosion, oxidation, contamination, or mechanical deformation, all of which can influence hydrogen diffusion and decrepitation behaviour.

The presence of unknown coating materials also represents a critical limitation for recycling process design. If coating composition and magnet design information are not

disclosed to recyclers, process optimization becomes more difficult, and treatment effectiveness may become inconsistent between material streams.

The results of this study further suggest that hydrogen decrepitation alone may not always be sufficient as a stand-alone disassembly method for embedded EV rotor magnets. Depending on magnet coating design, and surface accessibility, additional pre-treatment steps such as thermal treatment and robotic disassembly may be necessary before hydrogen-assisted recycling can be effectively applied.

Therefore, at the present moment, under the evaluated treatment conditions and constraints related to metallic coating and surface availability, HD is not a viable as an industrial scale stand-alone integrated disassembly and recycling method for embedded magnets.

5.3 Conditional recycling pathways for EoL Nd-Fe-B magnets from EV applications

5.3.1.1 Business decision-support frameworks

The proposed business decision framework presented in **Figure 19** and **Figure 20**, in combination with **Table 7,** demonstrates that the selection of recycling pathways for EoL EV magnets should not be based solely on environmental performance or circularity potential. Technical feasibility, material condition, feedstock availability, contamination, and recovery objectives must also be considered when evaluating recycling strategies.

The framework developed in this study integrates experimental feasibility observations under specific conditions with broader business and operational considerations for hydrogen-assisted recycling routes. This approach supports system-level evaluation of the magnet circularity while distinguishing between short-loop and long-loop recycling pathways. In addition, the highlighted features, advantages, and drawbacks from the routes explored in the decision frameworks, will support future Research & Development business activities that pursue sustainability of EoL EV materials, according to the company's and its partners' main recovery objectives and available EoL material sourcing.

The results of this work also suggest that magnet design and coating selection may strongly influence future recycling feasibility for embedded magnets. Previous studies have shown that magnets containing only polymeric coatings can undergo successful pulverization. Therefore, future design-for-disassembly approaches could evaluate the use of magnets without metallic alloy coatings in EV applications. The operational requirements and performance should be maintained while also enabling easier recycling and powder recovery. If technically viable, the use of more recycling-compatible magnet designs could improve hydrogen-assisted recovery efficiency and facilitate circularity of both the Nd-Fe-B magnets and the surrounding rotor laminated stack materials.

5.3.1.2 Basis for recycling pathway classification

The classifications presented in **Table 7** were developed though qualitative comparison of the recycling pathways identified in literature. Technology maturity was assessed based on reported industrial implementation, commercial availability, and demonstrated scalability. On the other hand, process complexity was evaluated considering the number of processing steps, operating conditions, reagent requirements, and separation stages involved in each route.

HD and HDDR methods were classified as highly mature technologies because both processes have been extensively investigated and are already applied for permanent magnet powder processing and low scale recycling. Similarly, hydro- and pyro- metallurgical routes, such as complete and selective leaching, and molten-salt electrolysis were classified as mature due to their widespread industrial use in rare-earth processing and recovery, including a variety of patented routes. Ionic-liquid extraction is classified as a low maturity technology as its efficacy is still under research at laboratory scale.

Process complexities were classified as low because all evaluated routes are based on established engineering principles and do not require highly specialized or emerging equipment. However, this classification should not be interpreted as low operational costs or low environmental impacts. For instance, hydrometallurgical routes often involve moderate to high energy and chemical reagent consumption and wastewater treatment requirements, despite being based on mature technologies, and are therefore classified as moderate in relation to their environmental perspective. Molten salt electrolysis, despite being a pyrometallurgical process that involves elevated temperature and moderate to high gas emissions, is considered more environmentally favourable than hydrometallurgical technologies. This is attributed to the reusable electrolytes and shorter processing chain, in addition to high quality of resulting material. On the other hand, although the technology is mature and established, usage as a REE recovery process still presents scalability challenges.

The classifications should therefore be interpreted as comparative indicators intended to support strategic decision-making rather than absolute performance metrics. A guide to classification interpretation is given in **Table 8**.

Table 8: Qualitative classification criteria used in the strategic assessment framework based on literature assessment.

Category	Low	Moderate	High
Circularity potential	Recovering of only element constituents or mixed products leading to substantial loss of material value	Recovery of REE that requires reprocessing before magnet remanufacturing	Preservation of the magnet alloy and/or properties, and viability of magnet remanufacturing with small loss of material value
Technology maturity	Laboratory scale, emerging technology	Pilot scale implementation, or presenting challenges to achieve industrial scale up	Commercially established process, demonstrated at industrial scale (even if for other processes)
Process complexity	Lower number of processing stages, easily controlled process	Multiple unit operations, easy-moderate to control process	Multiple separation/purification stages, need for specialized equipment/personnel, high process control requirements
Environmental favourability	High energy consumption, significant reagent requirements, and/or substantial waste/emission generation (Not favourable)	Moderate energy and reagent consumption, and/or moderate waste/emissions generation (Moderate to Favourable*)	Low energy demand, low reagent consumption, limited waste/emissions generation (Very favourable)

* A favourable environmental outlook is related to process optimization and high-quality recovery levels that that lead to lower environmental impacts, despite the route type presenting moderate resource consumption or emission levels. For instance, the molten-salt electrolysis, despite being a pyrometallurgical process which requires moderate-high energy consumption and moderate-high emission levels, requires less processing steps, and has lower wastewater generation in comparison to hydrometallurgical processes of similar consumption patterns.

5.3.2 Study limitations

The present study faced limitations related to sample availability, experimental conditions, and characterization scope. The experimental assessment was conducted using a limited number of samples, which restricts the representativeness of the results. Additionally, the analysed rotor samples were new rather than EoL, meaning they did not present typical degradation such as corrosion, contamination, oxidation, or mechanical deformation that would be expected in industrial recycling systems. Such conditions can greatly influence experimental results, including material quality and suitability for producing new sintered magnets.

The experimental procedures were performed in an external laboratory, which introduced constraints related to equipment availability and response time during testing. The available equipment at the beginning of this study did not allow pressure variation, limiting the scope of parameter assessment for initial samples. Furthermore, the available commercial equipment set up could not fit the full rotor sample, requiring it to be cut into smaller sections prior to the hydrogen treatment. While suitable for laboratory evaluation, this approach may not reflect the needs of an industrial-scale process.

Another limitation was the lack of detailed information regarding the presence and composition of metallic coatings on the embedded magnets due to supplier confidentiality. The existence of metallic coating layers hinders the hydrogen treatment at mild conditions – room temperature and atmospheric pressure – and might additionally influence the pulverization mechanism as it restrains volume expansion.

Finally, the limited project timeframe restricted extensive characterization of the resulting powder materials, particularly regarding detailed composition and magnetic property analysis, and restricted performing a full LCA of the Nd-Fe-B magnets.

Therefore, for the development of a reliable industrial-scale process, with HD as a disassembly and recovery method for EV magnets, it is necessary to have reliable data on the received materials' design and composition, and to conduct further comprehensive analysis for EoL samples.

6 Conclusion

This thesis investigated the potential of hydrogen decrepitation (HD) as a disassembly and recovery method for Nd-Fe-B permanent magnets embedded in electric vehicle (EV) rotors. The work combined experimental evaluation, environmental impacts and eco-costs assessment, and the creation of business-oriented decision frameworks to support the development of more circular recycling strategies for EV traction systems.

The environmental assessment highlighted the significant impacts associated with primary Nd-Fe-B magnet production, particularly due to the extraction and processing of REE, reflecting high associated eco-costs. The results also showed that recycled NdFeB magnets can substantially reduce the eco-costs when compared to virgin material production. At the same time, the study identified limitations and inconsistencies across currently available environmental databases and literature regarding Nd-Fe-B manufacturing routes, raw material sourcing, and process traceability. These limitations demonstrate the importance of improving transparency and standardization in environmental assessments and manufacturing routes of RE-based materials.

The experimental investigation demonstrated that hydrogen decrepitation of embedded EV rotor magnets may be technically possible under specific conditions. However, the results also showed that magnet accessibility, protective coatings, and equipment limitations strongly influence the effectiveness of the process. Samples with intact metallic coatings did not show any significant reaction, while partial decrepitation and hydrogenation were observed after complete removal of the exposed coating surface. XRD analysis confirmed the presence of hydrogenated phases, indicating hydrogen uptake occurred. Nevertheless, full pulverization was not achieved under the tested conditions, and further characterization is required to evaluate powder yield, material quality, and suitability for magnet-to-magnet recycling applications.

The study further showed that metallic coatings and limited surface exposure may hinder hydrogen diffusion into the magnet bulk, reducing decrepitation effectiveness. These findings suggest that HD alone may not be sufficient as a stand-alone disassembly strategy for embedded EV magnets. Instead, hybrid approaches combining HD with pre-treatment, thermal treatment, or mechanical and robotic disassembly may be required depending on rotor design and magnet configuration.

In addition to the experimental work, this thesis proposed a structured recycling route decision framework integrating technical feasibility, environmental considerations, and business-related factors. The framework supports evaluation of recycling pathways for EoL EV magnets while considering the business' material recovery objectives, process scalability, and circularity potential. This approach contributes to more informed decision-making for industrial implementation of RE magnet recycling strategies, and to further evaluation of the design-for-disassembly approach.

Overall, this work contributes to the ongoing research and development of circular solutions for Nd-Fe-B magnets in EV applications. By investigating the challenges associated with embedded magnet recovery, the thesis supports efforts to reduce supply-chain dependency on primary RE extraction, improve resource efficiency, and promote more sustainable management of critical raw materials in the future electrified mobility systems.

7 Future work recommendations

To further evaluate the implementation of disassembly and short-loop recycling of Nd-Fe-B magnets from EV applications, additional investigations are recommended. These studies would help reduce uncertainty related to magnet composition, coating behaviour, process scalability, and recovered material quality:

- Further HD experiments should be conducted using real end-of-life EV magnets, rather than only new rotor samples, to evaluate the effects of ageing, corrosion, coating degradation, adhesive condition, and material variability on hydrogen decrepitation performance.

- Commercial equipment constraints should be evaluated, including hydrogen handling, pressure control, temperature control, chamber volume, safety requirements, sample-size limitations, and scalability for industrial rotor components.

- Further characterization of recovered material is required to assess the viability of magnet-to-magnet recycling. This should include powder yield, particle-size distribution, oxygen content, phase composition by XRD, microstructure by SEM, chemical composition, and magnetic properties such as remanence, coercivity, and maximum energy product.

- The effect of surface availability should be studied systematically, including the role of adhesive bonding, limited exposed surface area, coating damage, and partial coating removal. Future studies should evaluate whether pressure variation under mild operating conditions can overcome these limitations without compromising yield, processing time, or material quality.

- HDDR should be evaluated as a possible post-treatment or alternative recycling route for embedded EV magnets. Its feasibility should be compared with HD in terms of powder quality, magnetic-property recovery, process complexity, energy demand, and cost-benefit performance.

- The technical viability of using magnets without metallic coating systems for high-performance EV rotor systems should be evaluated in combination with design-for-disassembly alternatives. This will increase the circularity potential of RE-based EV magnets.

- A material-flow analysis should be conducted for EV permanent magnet recovery in Sweden, including expected future EoL magnet volumes, collection logistics, recycling infrastructure, and the minimum material flow required for economically viable industrial-scale recycling.

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